

Portfolio Construction

A Plain-Language Study Guide

All six learning modules explained in everyday English, with the formulas you must memorise, the worked examples from the curriculum, and the traps that catch candidates in the exam.

- 1 Portfolio Risk and Return: Part I**
Diversification, utility, the efficient frontier
- 2 Portfolio Risk and Return: Part II**
CAPM, beta, and performance measurement
- 3 Portfolio Management: An Overview**
The process, investor types, the industry
- 4 Basics of Portfolio Planning & Construction**
The IPS, constraints, asset allocation
- 5 The Behavioral Biases of Individuals**
Why real investors act irrationally
- 6 Introduction to Risk Management**
Governance, risk budgeting, hedging

How to use this guide

Six modules, one continuous argument. Read it in order the first time.

Volume 9 looks like six separate topics. It is really one argument told in six steps, and it is much easier to remember once you can see the thread running through it:

1. Modules 1 and 2 build the theory. They answer one question: which risks does the market pay you to take, and which does it not? The answer - systematic risk is paid, non-systematic risk is not - is the foundation of everything after it.
2. Module 3 zooms out. It shows why you must think in portfolios rather than individual securities, and who the different investors are.
3. Module 4 makes it personal. It turns the theory into a written plan for one specific client: the investment policy statement.
4. Module 5 is the reality check. It explains why that client - and you - will quietly sabotage the plan through predictable psychological biases.
5. Module 6 gives you the machinery to hold the line: governance, risk budgets, measurement, and hedging tools.

KEY IDEA

Volume 9 in one sentence: risk is not the enemy - unmeasured, unchosen, unmanaged risk is. Every module is about choosing risk deliberately and then keeping the risk you actually have lined up with the risk you meant to take.

How the panels work

Three kinds of highlighted box appear throughout:

KEY IDEA

A blue KEY IDEA box holds a conclusion worth memorising outright. If you remember nothing else from a section, remember these.

EXAM TRAP

An orange EXAM TRAP box flags a place where the obvious answer is wrong, or where two similar-sounding concepts are routinely confused. These are where marks are actually lost.

Formulas sit in a grey monospaced box like this one.

A short note underneath explains what each symbol means and when to use it.

A note on the mathematics

You do not need to derive anything in Volume 9. You do need to be able to plug numbers into perhaps a dozen formulas quickly and correctly, and - more importantly - to say in words what each one means. Examiners test understanding far more often than arithmetic. Where a formula appears in this guide, the note underneath tells you what it is really saying.

One habit will save you marks repeatedly: check whether a question is asking about total risk (standard deviation) or systematic risk (beta). A large number of Volume 9 questions hinge on nothing more than that distinction.

LEARNING MODULE 1

Portfolio Risk and Return: Part I

How individual investments combine into portfolios, why correlation matters more than anything else, and how to find the one portfolio that is best for a particular investor.

1.1 Historical returns: what the record actually shows

Before any theory, the curriculum establishes a factual base. Over 1926-2017 in the United States, asset classes lined up almost exactly as risk-return theory would predict.

Asset class	Return	Risk (SD)
Small company stocks	12.1%	31.7%
Large company stocks	10.2%	19.8%
Long-term corporate bonds	6.1%	8.3%
Long-term government bonds	5.5%	9.9%
Treasury bills	3.4%	3.1%
Inflation	2.9%	4.0%

Read down the table and the pattern is unmistakable: every step up in return came with a step up in risk. That is the risk-return trade-off, and it is not a coincidence. It exists precisely because investors are risk-averse. If people did not dislike risk, risky assets would not have to offer more to attract buyers, and the two columns would not line up at all.

There is one instructive exception. Long-term government bonds carried more total risk (9.9%) than corporate bonds (8.3%) yet returned slightly less. This does not mean governments were more likely to default. It means government bonds happened to be more variable over this particular period, largely because of interest-rate movements. Real data is messy; the trend still holds.

Historical return is not expected return

The curriculum is emphatic about this. Historical return is what actually happened. Expected return is what a marginal investor requires going forward, and it is built from three pieces:

$$1 + E(R) = (1 + r_{rF}) \times [1 + E(\pi)] \times [1 + E(RP)]$$

r_{rF} = real risk-free rate (payment for postponing consumption); $E(\pi)$ = expected inflation; $E(RP)$ = expected risk premium.

In practice everyone uses historical averages as a proxy for expected returns, because there is nothing better. But it is an assumption, and the curriculum wants you to know it is an assumption. Large US stocks returned 18.2% in the 1990s and minus 1.0% in the 2000s. Whichever decade you had used to set expectations, you would have been badly wrong about the next one.

Real returns and the power of compounding

Because inflation ranged from minus 10.3% to plus 13.3%, comparing nominal returns across eras is misleading. In real terms, 1900-2017, one dollar grew to \$1,654 in equities, \$10.20 in bonds and \$2.60 in bills. The annual gap looks modest - 6.5% versus 2.0% versus 0.8% - but compounded over 118 years it is enormous. This is the single best argument in the curriculum for holding growth assets over long horizons.

1.2 Why the normal distribution is not enough

Mean and variance fully describe an investment only if returns are normally distributed and markets are efficient both informationally and operationally. Real returns violate this in two specific ways.

- Skewness. Returns are negatively (left) skewed: the distribution has a long tail of large losses. Crashes are bigger and faster than booms.
- Kurtosis (fat tails). Extreme outcomes of either sign happen far more often than a bell curve predicts. Because mean-variance analysis assumes they are rare, it systematically understates real risk.

The practical response is to supplement standard deviation with measures aimed at the tail, such as value at risk and conditional tail expectations - both of which return in Module 6. Market participants widely regard the underappreciation of tail events as a major contributing factor to the 2008 financial crisis.

Market characteristics: liquidity is a cost and a risk

Trading costs have three parts: brokerage commission, the bid-ask spread, and price impact. Liquidity affects the last two. The curriculum's arithmetic is worth remembering because it is so clean: a 10-cent spread on a \$100 stock is 0.1%, but the same 10-cent spread on a \$10 stock is 1%. The cheaper stock must earn 0.9% more just to break even against the dearer one.

Price impact is the second effect. Buying 100 shares of a liquid stock may move nothing; buying 100,000 shares may move the price a long way, because you have to keep raising your bid to persuade more holders to sell. Liquidity matters most in emerging markets, in corporate bonds, and for large institutions.

1.3 Risk aversion and utility

The curriculum's set-up: you may take GBP 50 with certainty, or a coin flip paying GBP 100 or nothing. Both have an expected value of GBP 50. Your choice reveals your type.

Choice	Type	What it implies
Takes the gamble	Risk seeking	Gains utility from uncertainty; would accept an expected GBP 45 gamble over a sure GBP 50
Genuinely indifferent	Risk neutral	Cares only about return, not risk
Takes the sure GBP 50	Risk averse	Might accept a sure GBP 45 to avoid the gamble entirely

The curriculum then makes a point that is easy to skim past but often tested: a single choice does not determine anybody's risk aversion. Buying a lottery ticket or visiting a casino does not make somebody a risk seeker in their investment portfolio. Because historical data shows a consistently positive risk-return relationship, we conclude that market prices are set by risk-averse investors, and every remaining assumption in the volume follows from that.

EXAM TRAP

Risk tolerance and risk aversion are opposites, not synonyms. Higher risk tolerance means lower risk aversion, which means a lower value of A in the utility function. Questions frequently swap the two words to see whether you are reading carefully.

The utility function

$$U = E(r) - \frac{1}{2} A \sigma^2$$

U = utility; E(r) = expected return; σ^2 = variance; A = risk aversion coefficient. Returns and standard deviations must be entered as decimals, not percentages.

Five properties the curriculum draws out, all of which have appeared as questions:

1. Utility is unbounded in both directions - it can be hugely positive or hugely negative.
2. Higher return raises utility; higher variance lowers it, and the reduction is amplified by A .
3. $A > 0$ for a risk-averse investor, $A = 0$ for a risk-neutral one, and $A < 0$ for a risk lover, for whom extra risk actually raises utility.
4. Utility only ranks investments. A portfolio scoring 4 is preferred to one scoring 2, but it is not 'twice as good'. There is no meaningful unit.
5. A risk-free asset has $\sigma^2 = 0$, so the second term vanishes and it produces identical utility for everyone regardless of A .

Worked example

Expected return 10%, standard deviation 20%, $A = 3$.

$$U = 0.10 - 0.5 \times 3 \times 0.20^2 = 0.10 - 0.06 = 0.04$$

The certainty equivalent is therefore 4%. To this investor, a risky 10% is worth exactly the same as a guaranteed 4%. The 6% gap is the price they place on the risk.

Indifference curves

An indifference curve joins every risk-return pair giving the same utility. For a risk-averse investor it slopes upward (more risk demands more return) and is convex, because each extra unit of risk requires an increasingly large compensation. Curves for the same investor can never touch or cross - if they did, the investor's preferences would be internally inconsistent.

- Steeper curve = more risk-averse. The most risk-averse investor has the steepest slope.
- Utility rises to the north-west: more return, less risk.
- Risk-neutral investors have flat, horizontal curves - risk simply does not enter their utility.
- Risk lovers have downward-sloping curves - they will give up return to obtain more risk.

1.4 Portfolio return and portfolio risk

This is the mathematical heart of Module 1, and the asymmetry between the two formulas is the whole point.

$$\text{Return: } R_p = w_1 R_1 + w_2 R_2 \quad (\text{a simple weighted average})$$

$$\begin{aligned} \text{Risk: } \sigma_p^2 &= w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2 \\ \sigma_p &= \sqrt{(w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2)} \end{aligned}$$

$$\text{Covariance: } \text{Cov}(R_1, R_2) = \rho_{12} \sigma_1 \sigma_2$$

Weights must sum to 1. Note the third term: it is the only place correlation enters, and it is where all diversification comes from.

KEY IDEA

Portfolio return is always a weighted average of the component returns. Portfolio risk is NOT a weighted average of the component risks - except in the single case where $\rho = +1$. That exception is the entire reason diversification works.

What correlation does, case by case

Correlation	Effect on portfolio risk	Interpretation
$\rho = +1$	$\sigma_p = w_1 \sigma_1 + w_2 \sigma_2$ (weighted average)	No diversification benefit at all
$\rho < +1$	$\sigma_p < \text{weighted average}$	Risk falls while return is unchanged
$\rho = 0$	Substantial reduction	Assets move independently
$\rho = -1$	Risk can be driven to exactly zero	A perfect hedge is possible

The curriculum demonstrates this with two identical stocks, each returning 10% with 20% risk, held 50/50. At $\rho = +1$ portfolio risk is 20% - no improvement. At $\rho = 0$ it falls to 14%. At $\rho = -1$ it falls to 0%. The expected return is 10% in all three cases. You have removed risk without giving up a single basis point of return.

The Beachwear example - why correlation, not count, is what matters

This is the clearest illustration in the volume. Beachwear earns 20% in a sunny year and 0% in a rainy one; average 10%, and risky.

- Add Snackshop, which also earns 20% when sunny and 0% when rainy. You now hold two businesses instead of one - but they are perfectly correlated ($\rho = +1$). The portfolio still earns 20% or 0%. Risk is completely unchanged.
- Add DVDrental instead, which earns 20% when rainy and 0% when sunny ($\rho = -1$). Now you earn exactly 10% whether it rains or shines. Risk has been eliminated and the return is untouched.

KEY IDEA

Adding more holdings does not diversify a portfolio. Adding holdings that behave differently does. This is why a portfolio of ten bank stocks is barely diversified at all.

1.5 Many assets: what survives diversification

Extend to N assets, equally weighted, with average variance and average covariance:

$$\sigma_p^2 = \sigma^2/N + [(N-1)/N] \times \text{Cov}$$

and if all variances and correlations are equal:

$$\sigma_p = \sqrt{(\sigma^2/N + [(N-1)/N] \rho \sigma^2)}$$

As N rises, the first term (individual variance) shrinks toward zero; the second term converges on the average covariance.

KEY IDEA

In a large portfolio, individual asset variances become irrelevant. Almost all remaining risk comes from how the assets move together. If assets were genuinely unrelated, portfolio risk could approach zero - but in reality most assets are positively correlated, so a floor of undiversifiable risk always remains. That floor is systematic risk, and Module 2 prices it.

Historical correlations confirm the difficulty. US large-cap versus US small-cap is about 0.72; US large-cap versus international is 0.66. Real diversification comes from bonds: US small-cap versus long-term Treasuries is actually negative at minus 0.15, and T-bills are close to zero against everything. Above 0.90 is considered high and offers little help; below 0.50 is genuinely useful.

Practical avenues for diversification

- Across asset classes - the biggest single lever available.
- Through index funds and ETFs - covering ten asset classes properly might need 300 securities, which is expensive to trade and track directly.
- Across countries - different industry mixes, policies and currencies. Currency returns are largely uncorrelated with stock returns, which helps even when the foreign market itself is risky.
- By NOT owning your employer's stock. You have already invested your human capital there. If the company fails you lose your job and your savings together.
- By evaluating each addition properly rather than accumulating holdings.
- By buying insurance or put options - assets with negative correlation and negative expected return that are still rational to hold because of the risk they remove.

The rule for adding a new asset

Add the asset if:

$$[E(R_{\text{new}}) - R_f] / \sigma_{\text{new}} > \{ [E(R_p) - R_f] / \sigma_p \} \times \rho_{\text{new},p}$$

In words: add the asset if its Sharpe ratio beats the existing portfolio's Sharpe ratio multiplied by its correlation with that portfolio. Low correlation lowers the hurdle - which is why a mediocre but uncorrelated asset can still improve a portfolio.

1.6 Frontiers, the CAL, and the optimal portfolio

The construction runs in a fixed order. Examiners test the order as often as the content.

1. Investment opportunity set - every portfolio that can be built from the available assets. Adding a new asset class that is not perfectly correlated pushes the whole set north-west, which is strictly better.
2. Minimum-variance frontier - for each level of return, the portfolio with the least risk. No risk-averse investor holds anything to the right of it.
3. Global minimum-variance portfolio - the leftmost point on that frontier. No portfolio of risky assets can have less risk than this one.
4. Markowitz efficient frontier - the portion of the minimum-variance frontier lying ABOVE the global minimum-variance portfolio. The part below is inefficient: for the same risk you could have had more return.
5. Add the risk-free asset. Lines from R_f to points on the efficient frontier are capital allocation lines. The steepest one - the tangent - dominates all others, and its tangency point is the optimal risky portfolio, P.
6. Overlay the investor's indifference curves. The point where the highest attainable curve touches the CAL is that investor's optimal portfolio.

$$E(R_p) = R_f + [(E(R_i) - R_f) / \sigma_i] \times \sigma_p$$

Intercept = R_f . Slope = the market price of risk: the extra return demanded for each additional unit of risk.

Note the slope of the efficient frontier flattens as you move right. Moving from the global minimum-variance portfolio to point A might gain the same return as moving from A to D, but the extra risk taken in the second step is three to four times larger. Investors receive decreasing increases in return as they assume more risk.

The two-fund separation theorem

KEY IDEA

Every investor, regardless of wealth, taste or risk aversion, holds a combination of just two things: the risk-free asset and the SAME optimal risky portfolio P. Only the proportions differ.

This splits the problem into two independent decisions. The investment decision - identifying P - is purely technical and identical for everybody. The financing decision - how much to lend or borrow - is where personal risk preference enters. Points left of P are lending portfolios; points

right of P are borrowing (leveraged) portfolios, achieved by borrowing at R_f and investing more than 100% in P.

EXAM TRAP

A common error: assuming a highly risk-averse investor holds a different, safer risky portfolio. They do not. They hold the same portfolio P, just less of it, with the rest in the risk-free asset.

Comprehensive worked example (the Lohrmanns)

Two risky assets: A returns 20% with 50% risk; B returns 15% with 33% risk; correlation zero. A risk-free asset returns 3%. The slope-maximising weight turns out to be 38.20% in A, which produces the capital allocation line:

$$E(R_p) = 0.03 + 0.4978 \sigma_p$$

0.4978 is the market price of risk - each extra 1% of standard deviation buys about 0.50% of extra expected return.

With a risk aversion coefficient of 2.5, computing utility at several points along this line gives 0.0300 at a 3% return, 0.0717 at 9%, 0.0774 at 15% and 0.0546 at 20%. Utility rises and then falls, so the Lohrmanns' optimal portfolio is the 15% return / 24.1% risk point.

One detail worth noticing: on the CAL a 20% return can be obtained with 34.2% risk, whereas holding Asset A alone gives the same 20% return with 50% risk. Combining assets with the risk-free asset is simply a better deal than holding any single risky asset.

LEARNING MODULE 2

Portfolio Risk and Return: Part II

The Capital Asset Pricing Model: which risk gets paid, how beta measures it, and how to judge whether a manager actually added value.

2.1 From the CAL to the Capital Market Line

The capital market line is a special case of the capital allocation line - the one where the risky portfolio is the market portfolio itself.

$$E(R_p) = R_f + [(E(R_m) - R_f) / \sigma_m] \times \sigma_p$$

The x-axis is total risk (standard deviation). The CML applies only to efficient, well-diversified portfolios.

Points above the CML are unattainable; points below it are dominated. Note that although the CML combines two assets, it is a straight line - unlike a combination of two risky assets, which curves. The reason is arithmetic: the risk-free asset has zero standard deviation and zero covariance with everything, so two of the three terms in the portfolio variance formula disappear and σ_p reduces to $(1 - w_1)\sigma_m$.

A worked case

T-bills at 5%; the market at 15% return with 20% risk. The CML is therefore $E(R_p) = 5\% + 0.50 \sigma_p$.

% in market	Expected return	Risk
0% (all T-bills)	5.0%	0%
25%	7.5%	5%
75%	12.5%	15%
100%	15.0%	20%
125% (borrow 25%)	17.5%	25%
200% (borrow 100%)	25.0%	40%

Leverage extends the line to the right. Negative investment in the risk-free asset - borrowing - raises both expected return and risk proportionally.

When borrowing costs more than lending

Realistically you cannot borrow at the government's rate. If you lend at R_f but borrow at the higher R_b , the CML kinks at the market portfolio: it keeps the slope $(R_m - R_f)/\sigma_m$ up to M, then flattens to $(R_m - R_b)/\sigma_m$ beyond it. In the curriculum's example the slope drops from 0.50 to 0.40 - each 1% of extra risk now buys only 0.40% of extra return. Leverage still works, but on worse terms, so only less risk-averse investors will choose it.

Does a single optimal risky portfolio really exist?

Only under homogeneity of expectations - the assumption that everyone analyses securities identically and reaches identical valuations. The curriculum's Siemens illustration shows how fragile that is: the shares closed at EUR 111.84, but one analyst might value them at EUR 95 (overvalued, weight zero) and another at EUR 125 (undervalued, overweight). Different valuations mean different optimal portfolios.

The saving argument is that the market price reflects what the marginal, informed investor believes, so the market portfolio remains the natural benchmark against which everything else is judged - even if no individual actually holds it.

This distinction also defines passive versus active management. Passive investors accept market prices as unbiased and track an index cheaply. Active investors trust their own estimates, overweighting what they think is cheap, underweighting or shorting what they think is dear.

EXAM TRAP

What is 'the market'? Theoretically every risky asset with value, including human capital and the Taj Mahal. In practice a proxy such as the S&P 500 is used - which covers roughly 80% of US equity capitalisation but only about 32% of world equity, and no bonds or property. Every CAPM result inherits that approximation. Roll's critique says this is why CAPM cannot truly be tested.

2.2 Systematic and non-systematic risk

	Systematic risk	Non-systematic risk
Also called	Market, non-diversifiable	Company-specific, idiosyncratic, diversifiable
Sources	Interest rates, inflation, economic cycles, political uncertainty, widespread natural disaster	A failed drug trial, an airliner crash, a management scandal
Diversifiable?	No	Yes
Is it paid?	Yes	No

Total variance = Systematic variance + Non-systematic variance

$$\sigma_i^2 = \beta_i^2 \sigma_m^2 + \sigma_e^2$$

$$\sigma_i = \sqrt{(\beta_i^2 \sigma_m^2 + \sigma_e^2)}$$

The components add as VARIANCES, never as standard deviations. This is a classic exam trap.

Why non-systematic risk earns nothing

Follow the argument through, because it is examinable as reasoning rather than as a fact to recall. Suppose diversifiable risk did pay a premium. You would buy assets loaded with it, then

diversify it away by combining uncorrelated holdings. You would end up bearing only systematic risk while still collecting payment for non-systematic risk you no longer hold - a free lunch.

Every investor would do the same. Demand for such assets would rise, their prices would climb, and their expected returns would fall until the free lunch disappeared. Equilibrium is therefore reached only when the premium for diversifiable risk is exactly zero.

KEY IDEA

Only risk that cannot be avoided is compensated. This is the single most important conclusion in Module 2, and everything downstream - beta, the CAPM, the SML, the Treynor ratio, Jensen's alpha - is a consequence of it.

A caveat the curriculum adds: this does not mean non-systematic risk is harmless. It matters greatly to company management, to poorly diversified investors, and to analysts covering individual firms. And what looks like company-specific risk can turn out to be systematic - poor credit management at one bank became a global crisis because the bank was too big to fail.

2.3 Return-generating models and beta

Building a true market portfolio is impractical - 1,000 assets would require 1,000 return estimates, 1,000 standard deviations and 499,500 correlations. Return-generating models cut through this by estimating expected return from a small number of factors.

- Macroeconomic factor models - growth, interest rates, inflation, employment, consumer confidence.
- Fundamental factor models - earnings, growth, cash flow, R&D, patents.
- Statistical factor models - factors extracted from the data itself. The curriculum warns these can produce nonsense: the conference of the Super Bowl winner appears to explain US stock returns, with no economic link whatsoever. Data mining generates spurious factors, so analysts prefer the first two types.

The simplest version is the single-index model, $E(R_i) - R_f = \beta_i[E(R_m) - R_f]$. Its practical implementation is the market model, estimated by regression:

$$R_i = \alpha_i + \beta_i R_m + e_i$$

α is the intercept, β the slope, e the company-specific error. Fitted by regressing historical security returns on market returns.

A worked case: Wal-Mart's regression gives $\alpha = 0.0001$ and $\beta = 0.90$. On a day when the market rises 1%, the expected return is $0.0001 + 0.90 \times 0.01 = 0.91\%$. If Wal-Mart actually rose 2%, the abnormal (company-specific) return is $2\% - 0.91\% = 1.09\%$.

Calculating beta

$$\beta_i = \text{Cov}(R_i, R_m) / \sigma_m^2 = \rho_{i,m} \times (\sigma_i / \sigma_m)$$

Two equivalent forms. Use the first when given covariance, the second when given correlation and standard deviations.

- Market beta = 1 by definition (its correlation with itself is 1).
- Risk-free asset beta = 0 (zero covariance with everything).
- The average beta of all assets in the market is 1.
- Positive beta means the asset moves with the market; negative beta means against it. Consistently negative betas are rare, because most stocks in developed markets correlate above 0.70 with the market.

Quick examples: correlation 0.70, asset SD 0.25, market SD 0.15 gives $\beta = 0.70 \times 0.25 / 0.15 = 1.17$. Gold with a large SD but zero correlation has $\beta = 0$. An IPO with SD 40% and correlation 0.7 against a 25% market has $\beta = 1.12$ - risky in total terms, but only modestly so systematically.

EXAM TRAP

High total risk does not imply high beta. Gold in the curriculum's example has the same standard deviation as the market but a beta of zero, because it is uncorrelated. Questions exploit this constantly.

On estimation: short windows (say 12 months) reflect current risk but are noisy and distorted by one-off events. Longer windows of three to five years are statistically steadier but may describe a company that no longer exists in that form. A three-year beta will differ from a five-year beta, and a beta from daily data will differ from one built on monthly data.

2.4 The CAPM and the Security Market Line

$$E(R_i) = R_f + \beta_i [E(R_m) - R_f]$$

The market risk premium is $E(R_m) - R_f$, also called the excess market return. Portfolio beta is simply the weighted average of component betas: $\beta_p = \sum w_i \beta_i$.

Two assets with the same beta have the same expected return, whatever they are - a bank, a gold mine, an emerging market fund. Under CAPM, beta is the only thing that matters. Assets with $\beta > 1$ are expected to beat the market; $\beta < 1$ to lag it; $\beta < 0$ to return LESS than the risk-free rate.

That last case is not a mistake. Insurance is the curriculum's example: it pays out precisely when your wealth is destroyed, so it reduces total portfolio risk and is valuable despite a negative expected return.

The six assumptions

1. Investors are risk-averse, utility-maximising and rational. They need not share the same degree of risk aversion, only the direction.
2. Markets are frictionless - no transaction costs, no taxes, no restrictions on short selling, and borrowing and lending at R_f are both possible.
3. All investors plan for the same single holding period.
4. Investors have homogeneous expectations - the assumption that produces one market portfolio and one security market line.
5. All investments are infinitely divisible.
6. Investors are price takers; no single trader moves prices.

The purpose of all six is to create a marginal investor who chooses a mean-variance efficient portfolio predictably. Relaxing most of them changes little. The exception the curriculum highlights is short selling: costs or restrictions on shorting can introduce an upward bias in asset prices and do genuinely threaten CAPM's conclusions.

SML versus CML - a distinction that is heavily tested

	Capital Market Line	Security Market Line
X-axis	Total risk (σ)	Systematic risk (β)
Applies to	Efficient, well-diversified portfolios only	ANY security or portfolio, efficient or not
Slope	$(R_m - R_f)/\sigma_m$, the market price of risk	$R_m - R_f$, the market risk premium
Used for	Asset allocation between R_f and the market	Pricing securities; spotting mis-valuation

KEY IDEA

Total risk and systematic risk are equal ONLY for efficient portfolios, because those have no diversifiable risk left. For an individual security they differ, which is exactly why the SML - and not the CML - is the tool for pricing single securities.

Worked examples

Risk-free 3%, market 13% with 23% risk. Bajaj Auto has 50% standard deviation but zero correlation with the market, so $\beta = 0$ and $E(R) = 3\%$ - the risk-free rate, despite being a volatile stock. Mueller Metals also has 50% risk but correlation 0.65, giving $\beta = 1.41$ and $E(R) = 3\% + 1.41 \times 10\% = 17.1\%$.

Portfolio example: 20% risk-free, 30% market, 50% in a stock with $\beta = 2.0$. $\beta_p = (0.20 \times 0) + (0.30 \times 1.0) + (0.50 \times 2.0) = 1.30$, so $E(R_p) = 4\% + 1.30 \times 12\% = 19.6\%$.

Using CAPM in capital budgeting

The GlaxoSmithKline case shows the model doing real work. Beta 2.3, risk-free 2%, market 12%, so the required return is $2\% + 2.3 \times 10\% = 25\%$. Discounting the probability-weighted cash flows at 25% gives an NPV of minus \$147.07 million, so the project is rejected. A lower discount rate would have approved it - which is precisely why the risk adjustment matters.

2.5 Where CAPM breaks down

Theoretical limitations

- Single-factor: only beta is priced, so no other characteristic may be considered. Simple and prescriptive, but restrictive and inflexible.
- Single-period: it cannot capture factors that vary over time or span several periods, which can encourage myopic decisions.

Practical limitations

- The true market portfolio includes non-investable assets such as human capital and is therefore unobservable - Roll's critique.

- Different proxies give different answers for the same asset, which the model does not permit.
- Beta estimates are unstable across periods and data frequencies.
- It predicts returns poorly. Empirical support is weak; realised returns are not determined by systematic risk alone.
- Homogeneous expectations plainly do not hold, so there is no unique optimal portfolio or single SML.

What came next

Arbitrage Pricing Theory allows many factors, and the factors need not be common across assets. It is theoretically elegant and more flexible than CAPM, but it does not tell you what the factors are - which is why CAPM remains preferred in practice.

Fama-French-Carhart four-factor model:

$$E(R_{it}) = \alpha_i + \beta_{MKT} \cdot MKT + \beta_{SMB} \cdot SMB + \beta_{HML} \cdot HML + \beta_{UMD} \cdot UMD$$

MKT = excess market return; SMB = small minus big (size); HML = high minus low book-to-market (value versus growth); UMD = up minus down (momentum).

Historically the coefficient on MKT is not significantly different from zero - an uncomfortable result, since it implies stock returns are largely unrelated to the market factor. Size, value and momentum do the explanatory work. The model predicts far better than CAPM, but two caveats apply: it has no equilibrium theory behind it, and there is no assurance it will keep working.

2.6 Performance appraisal

Performance evaluation answers three separate questions. Performance MEASUREMENT asks what the return and risk were. Performance ATTRIBUTION asks where the performance came from. Performance APPRAISAL asks whether it was skill or luck. The four ratios below belong to appraisal.

Measure	Formula	Risk used	Key property
Sharpe ratio	$(R_p - R_f) / \sigma_p$	Total	Slope of the CAL; must be compared with another portfolio to mean anything
Treynor ratio	$(R_p - R_f) / \beta_p$	Systematic	Breaks down for negative-beta assets
M ²	$(R_p - R_f)(\sigma_m / \sigma_p) + R_f$	Total	Sharpe rescaled into a percentage return; ranks identically to Sharpe
Jensen's alpha	$R_p - [R_f + \beta_p(R_m - R_f)]$	Systematic	Meaningful on its own; the maximum you should pay a manager

Both Sharpe and Treynor require a positive numerator to be meaningful. If the excess return is negative, a riskier portfolio produces a less negative ratio and the rankings invert - the measure becomes actively misleading.

M^2 answers the question 'what would this portfolio have returned if it had been levered or de-levered to exactly the market's volatility?' The gap between that figure and the market return is M^2 alpha. Worked example: $R_f = 4\%$, $R_p = 14\%$, $\sigma_p = 25\%$, $\sigma_m = 20\%$. Sharpe = 0.40, so $M^2 = 0.40 \times 0.20 + 0.04 = 12.0\%$. If the market returned 10%, the portfolio beat it by 2.0% on a risk-adjusted basis.

The three-manager case - and why it matters

Manager	Return	σ	β	Sharpe	Treynor	M^2 alpha	Alpha
X	10.0%	20.0%	1.10	0.35	0.064	0.65%	0.40%
Y	11.0%	10.0%	0.70	0.80	0.114	9.20%	3.80%
Z	12.0%	25.0%	0.60	0.36	0.150	0.84%	5.40%
Market	9.0%	19.0%	1.00	0.32	0.060	0.00%	0.00%

All three managers beat the market on every measure, so the fund should be pleased. But Y and Z swap places depending on the measure used. On total-risk measures Y wins decisively (Sharpe 0.80 versus 0.36). On systematic-risk measures Z wins (Treynor 0.150 versus 0.114; alpha 5.40% versus 3.80%). Manager X is worst on every measure.

KEY IDEA

Choose the measure to match the situation. If the client's whole wealth is in this one portfolio, total risk is what they experience - use Sharpe or M^2 , and hire Manager Y. If this is one sleeve of an already well-diversified fund, only systematic risk matters - use Treynor or Jensen's alpha, and hire Manager Z.

Note also that the risk-free asset has no Sharpe, Treynor or M^2 at all - each divides by a risk measure that is zero - but its alpha is defined and equals zero. The market's alpha and M^2 alpha are zero by construction.

Security selection with the SML

Plot each security's estimated return against its beta. Points on the line are fairly valued. Points ABOVE the line offer more return than their risk warrants - undervalued, buy. Points BELOW are overvalued - avoid or short.

The security characteristic line makes the same point in regression form: plot the security's excess return against the market's excess return, and the intercept is Jensen's alpha while the slope is beta.

Positive alpha does not always mean mispricing. A security can have positive alpha simply because it is poorly correlated with your benchmark and its return is generous for the systematic risk it adds. For portfolio construction, weight each non-market security in proportion to α_i/σ_{ei}^2 - more weight for higher alpha, less for higher idiosyncratic risk. The information ratio α_i/σ_{ei} measures abnormal return per unit of added risk; higher is better.

Practical worked case: three stocks outside the Nikkei 225. P has alpha of minus 0.02 and is excluded. Q has alpha 0.04 with non-systematic variance 0.0158, giving a relative weight of 2.53. R has alpha 0.03 with variance 0.0137, giving 2.19. Q therefore gets about 15.5% more

weight than R.

Finally, on diversification: roughly 30 randomly selected securities across different asset classes remove most non-systematic risk. If the holdings are not randomly selected - thirty banks, say - the reduction is far smaller, and an index fund would serve better.

LEARNING MODULE 3

Portfolio Management: An Overview

Why the portfolio - not the security - is the right unit of analysis, the three-step management process, who the investors are, and what the industry sells them.

3.1 The case for a portfolio approach

The whole volume rests on one decision: do you evaluate securities in isolation, or by what they contribute to the portfolio as a whole? The curriculum makes the case with a real disaster rather than with algebra.

Enron

Through the 1990s Enron returned over 27% a year against 13% for the S&P 500 and was one of the most admired companies in America. Employees held it through their 401(k). By January 2001 those accounts were worth over USD 2 billion, of which USD 1.3 billion - 62% - was Enron stock.

The detail that matters: only about USD 150 million of that USD 1.3 billion was actually restricted from sale. Employees were free to sell the rest and chose not to. One 67-year-old retiree held all USD 2 million of his retirement savings in Enron shares. Between January 2001 and January 2002 the price went from about USD 90 to zero.

KEY IDEA

The loss was not simply large - it was correlated with everything else these employees owned. Their wages, their job security and their savings all depended on the same company. When Enron failed they lost their income and their retirement at the same moment. The same pattern appeared at Owens Corning, Northern Telecom, Corning and ADC Telecommunications, where employees held more than 25% of assets in employer stock while prices fell almost 90%.

Portfolios reduce risk more than they change returns

The Hong Kong example quantifies it. Five HKSE-listed shares over 16 quarters: picking one at random gave an average return of 15.1% with 24.9% risk. An equally weighted portfolio of all five gave the same 15.1% return with only 17.9% risk.

$$\text{Diversification ratio} = 17.9\% / 24.9\% = 71.9\%$$

The portfolio's risk is about 72% of the average individual stock's risk. Lower is better. And this was achieved with five companies from a similar industry group - across genuinely different industries the ratio would have been lower still.

Composition also matters, not just the act of diversifying. The equally weighted portfolio gave 15.1% return at 17.9% risk. An optimised mix - 25% Yue Yuen, 3% Cathay Pacific, 52% Hutchison Whampoa, 20% Li & Fung, 0% COSCO - gave the same 15.1% return at just 15.6% risk.

The honest caveat: diversification fails when you need it most

Period	Mean return	Diversification ratio
Q4 1993 - Q3 2000	12.6%	75.1%
Q1 2006 - Q1 2009	-4.7%	95.8%
Q4 2007 - Q1 2009	-48.5%	99.4%

Read the third row carefully. A ratio of 99.4% means the diversified portfolio was almost exactly as risky as a single randomly chosen index. Correlations converged on one and every market fell together. Diversification did protect against picking the very worst market, but that was nearly all it did.

EXAM TRAP

The examinable conclusion: portfolios are most likely to provide RISK REDUCTION. They do not guarantee downside protection and they do not eliminate risk. In the face of a worldwide contagion, the curriculum says flatly, diversification is a false promise.

Modern portfolio theory dates from Markowitz's 1952 article. Its central conclusion is not merely 'hold portfolios' but 'focus on how the holdings relate to one another'. Sharpe, Lintner and Treynor then established that only the non-diversifiable portion of an asset's risk should affect its price - the insight that becomes the CAPM.

3.2 The three-step process

1. **PLANNING.** Understand the client's objectives and constraints, and write the investment policy statement. It may name a benchmark for later evaluation, and should be reviewed roughly every three years or whenever circumstances change materially.
2. **EXECUTION.** First set the target asset allocation, then analyse securities, then construct the portfolio, then trade. Analysis may be top-down (macro to industry to company) or bottom-up (company specifics first, less concerned with the economic cycle).
3. **FEEDBACK.** Monitor and rebalance as weights drift or the client's needs change; measure performance against the benchmark and report it. Findings may send you back to revise the IPS.

KEY IDEA

Order matters and is examined directly. Objectives and constraints are set in PLANNING. Asset allocation and security selection happen in EXECUTION. Rebalancing and performance reporting happen in FEEDBACK. Of all the decisions taken, asset allocation is commonly viewed as having the greatest impact on performance.

Two portfolios, same theory, opposite answers

Asset class	Yale endowment	MassMutual (life insurer)
Public equity	19.1%	9% (preferred + common)
Fixed income	4.6%	56% bonds
Private equity	14.2%	5% partnerships
Real assets / real estate	18.7%	1% + 14% mortgages
Absolute return / hedge funds	25.1%	—
Cash	1.2%	2%
Other	17.2%	13% (policy loans, other)

Yale holds under 5% in fixed income; MassMutual holds about 80% in bonds, mortgages, loans and cash. Neither is wrong. Yale has a perpetual horizon and no fixed liabilities; the insurer must pay unpredictable claims and faces regulatory constraints. Identical theory, opposite portfolios, because the constraints differ. This is the best single argument in Module 3 for why the IPS must come first.

3.3 Types of investor

Individuals and pension plans

	Defined contribution (DC)	Defined benefit (DB)
What is fixed	The contribution going in	The benefit coming out
Who bears investment risk	THE EMPLOYEE	THE EMPLOYER
Who bears inflation risk	The employee	The employer
Examples	401(k) US, group personal pension UK, superannuation Australia	Traditional corporate and public pension schemes
Trend	Growing	Shrinking - lower cost and risk to the company

Global pension assets exceeded USD 41 trillion at the end of 2017, with the US, UK and Japan accounting for over 76%. The DC/DB split varies sharply by country: Australia is 87% DC and the US 60% DC, while Japan is 96% DB, Canada 95%, the Netherlands 94% and the UK 82%.

EXAM TRAP

If a trustee has a fiduciary duty to beneficiaries and the sponsor's interests conflict with theirs - the sponsor wanting more risk to lower future funding costs, the beneficiaries wanting safety - the trustee must act in the BENEFICIARIES' best interests.

Institutional investors compared

Investor	Horizon	Risk tolerance	Income need	Liquidity need
Individuals	Varies	Varies	Varies	Varies
DB pension plans	Long	Quite high	High if mature, low if growing	Varies with maturity
Endowments & foundations	Very long (perpetual)	High	To meet spending rule	Quite low
Banks	Short	Quite low	To pay depositors	HIGH
Insurance companies	Short (P&C) / long (life)	Quite low	Typically low	High - to meet claims
Investment companies	Varies by fund	Varies	Varies	High - to meet redemptions
Sovereign wealth funds	Varies	Varies	Varies	Varies

Endowments and foundations differ in kind: endowments fund the operations of a non-profit institution, while foundations make grants. US endowments hold about 54% in alternatives, reflecting their perpetual horizon and the influence of the Yale model. Yale's spending rule illustrates the balancing act: a 5.25% long-term target with smoothing, so that spending in any year is 80% of last year's spending plus 20% of the target rate applied to the market value two years prior. The Wellcome Trust instead targets a 4.5% real return.

Insurance companies split their assets in two. The GENERAL account holds premiums and is invested conservatively in fixed income under regulatory guidance. The SURPLUS account is the excess of assets over liabilities and can target higher returns in equities, real estate, infrastructure and hedge funds. Life insurers hold longer assets than property and casualty insurers because their liabilities are longer and far more predictable.

3.4 The asset management industry

About USD 79 trillion under management at the end of 2017, with roughly 80% in North America and Europe, but the fastest growth in Asia and Latin America. A manager is a BUY-SIDE firm because it buys research and execution from SELL-SIDE broker-dealers.

Category	Assets	Revenue	% of assets	% of revenue
Actively managed	\$64trn	\$258bn	80%	94%
of which alternatives	\$12trn	\$117bn	15%	43%
Passively managed	\$16trn	\$17bn	20%	6%

KEY IDEA

The asset and revenue columns tell two completely different stories. Passive management holds a fifth of the world's assets but earns only 6% of the industry's revenue, because its fees are a fraction of active fees. Alternatives are the mirror image: 15% of assets producing 43% of revenue, because they charge both management and performance fees.

SMART BETA sits between the two: simple, transparent, rules-based strategies tilted toward factors such as size, value, momentum or dividends, with fees and turnover higher than plain market-cap indexing but below full active management.

Three industry trends

- Growth of passive investing. Concentrated among a few providers - the top three ETF managers hold about 70% of ETF assets. Driven by low cost and by the difficulty of generating alpha in efficient markets such as US large-cap.
- Big data. Social media sentiment, satellite imagery and geolocation data - retailer car-park occupancy, cargo ship movements - analysed with machine learning. An information arms race requiring heavy investment in data scientists and infrastructure.
- Robo-advisers. About USD 180 billion by end-2017. Roughly 0.20% a year against about 1% for a traditional adviser, serving younger and mass-affluent clients who were previously underserved.

3.5 Pooled investment products

A mutual fund is a commingled pool in which each investor has a pro-rata claim on income and value. Net asset value is computed daily from closing prices.

Open-end versus closed-end

	Open-end fund	Closed-end fund
New money	Accepted; new shares created at NAV	Not accepted; fixed share count
Trades at	Exactly NAV	Premium or discount to NAV
Growth	Easy	Limited
Main drawback	Must hold cash and may be forced to sell assets to meet redemptions	Cannot easily grow
Share of US MF assets	About 99%	About 1%

The curriculum's worked example: a fund starts at USD 10 million with 100,000 shares at USD 100. Later NAV per share is USD 120. Investor F invests USD 960,000, creating 8,000 new shares; Investor E redeems 5,000 shares for USD 600,000. Net effect: USD 360,000 of new money and 3,000 new shares. Every other investor's holding and share value is completely unaffected.

Separately, LOAD funds charge a percentage fee on purchase or redemption on top of the annual fee and are usually sold through brokers who take part of it. NO-LOAD funds charge only

the annual fee. Load funds have declined in importance.

Fund types by underlying asset

- Money market funds - very short instruments; constant or variable NAV; NOT insured like bank deposits, despite being used as a substitute for them.
- Bond funds - maturities from one year to thirty-plus, versus overnight to ninety days for money market funds. Categories include global, government, corporate, high yield, inflation-protected and national tax-free.
- Stock funds - historically the largest category. Active funds charge more and trade more, which in many jurisdictions realises more taxable capital gains; index funds are cheaper and buy-and-hold. The first index fund was launched by Vanguard in 1976.
- Hybrid / balanced funds - both stocks and bonds. More common in Europe than the US, and growing through lifecycle or target-date funds that automatically shift from equities to bonds as retirement approaches.

Other products

Product	What distinguishes it
Separately managed account (SMA)	You own the assets directly, not a share of a pool. Fully customisable for tax position and ESG exclusions. Drawback: a much higher minimum investment.
Exchange-traded fund (ETF)	Trades intraday on an exchange like a stock; can be shorted or bought on margin; dividends are paid out rather than reinvested; low minimum; NO capital gain distributions. Grew from \$428bn in 2005 to \$4.9trn by mid-2018.
Hedge fund	Short selling, leverage, absolute return objective, low correlation, and a two-part fee - traditionally 2% management plus up to 20% incentive, subject to a high-water mark. High minimums and restricted liquidity.
Private equity / venture capital	7-10 year life; hands-on involvement in portfolio companies; exit by merger, sale or IPO. Fees: management, transaction, carried interest (typically 20%, usually only after LPs recover their capital), and investment income.

EXAM TRAP

Buyout funds make a FEW LARGE investments in private companies expecting most to work. Venture capital funds make MANY SMALL investments expecting most to fail and a few to pay off spectacularly. Also remember: ETFs are the product least likely to make a capital gain distribution, and hedge funds are the pooled vehicle subject to the LEAST regulation.

LEARNING MODULE 4

Basics of Portfolio Planning and Construction

Turning a client's life into a written plan: the investment policy statement, the five constraints, and how the strategic asset allocation is built from it.

4.1 Why the IPS exists

Portfolio planning is the programme developed BEFORE constructing a portfolio that defines the client's investment objectives. The investment policy statement is the written document that governs it. It is the starting point of the entire management process, and without a full understanding of the client's situation success is unlikely.

Success here has a specific meaning: the client achieves their important goals by means they are comfortable with. Both halves matter. A plan that reaches the target through risks the client cannot stomach will be abandoned halfway - usually at the worst possible moment.

In several jurisdictions an IPS or equivalent is a legal requirement. UK pension schemes must have a statement of investment principles under the Pensions Act 1995; the UK regulator suggests reviewing it at least every three years. MiFID requires firms to categorise clients as eligible counterparties, institutional or retail, which determines the protections they receive.

EXAM TRAP

A written IPS is best practice and is sometimes legally required, but it does NOT ensure that objectives will be achieved. The examinable description is that it communicates a plan for TRYING to achieve investment success, and that it works best as a COLLABORATIVE effort between client and manager.

Structure of an IPS

- Introduction - describes the client
- Statement of purpose
- Statement of duties and responsibilities - client, custodian, managers
- Procedures - keeping the IPS current and handling contingencies
- INVESTMENT OBJECTIVES - risk and return
- INVESTMENT CONSTRAINTS - the five categories below
- Investment guidelines - how policy is executed: leverage, derivatives, excluded assets
- Evaluation and review
- Appendices - strategic asset allocation and rebalancing policy

KEY IDEA

The objectives and constraints sections are the ones most closely linked to the client's distinctive needs. The strategic asset allocation and rebalancing policy are typically APPENDICES. The statement of duties and responsibilities is an integral part of the main document, not an appendix - a distinction that has been tested directly.

4.2 Risk objectives

- **ABSOLUTE** - self-standing, unrelated to any market. 'Do not lose more than 4% of capital in any 12 months.' Better expressed as a probability: 'with 95% probability, do not lose more than 4% in any 12 months.' Measured by standard deviation, variance, or value at risk.
- **RELATIVE** - measured against a benchmark. 'Stay within 4% of the TOPIX return.' Measured by tracking risk, also called tracking error - the standard deviation of the differences between portfolio and benchmark returns.
- **LIABILITY-DRIVEN (LDI)** - where the size and timing of future obligations are known. A pension plan's risk objective becomes minimising the probability of failing to meet payments as they fall due.

A note on interpreting tracking error: because it is a one standard deviation measure, an expected tracking error of 2% means returns fall within roughly 4% of the index about 95% of the time, assuming normality.

4.3 Risk tolerance: ability versus willingness

	Ability BELOW average	Ability ABOVE average
Willingness BELOW average	Below-average risk tolerance	Conflict - resolution needed
Willingness ABOVE average	Conflict - resolution needed	Above-average risk tolerance

- **ABILITY** to bear risk is **OBJECTIVE**. Time horizon, stability and level of income, and wealth relative to liabilities. A 20-year horizon allows more scope to recover from losses than a 2-year horizon. Assets comfortably exceeding liabilities means high ability.
- **WILLINGNESS** to take risk is **SUBJECTIVE** - psychology, self-esteem, independence of thought, financial knowledge and decision style. Assessed by discussion or by a psychometric questionnaire such as Grable and Joo's five-item instrument, scored 5 to 20, where the sample mean was 12.86 and the median 13.

KEY IDEA

When ability and willingness conflict, adopt the **LOWER** of the two and document the decision. You may educate a client whose willingness rests on a miscalculation or misperception. You must **NOT** try to change a client's personality - modifying elements of personality is not within the investment adviser's role.

The two Gascon cases

	Marie Gascon	Jacques Gascon
Income	EUR 250,000, stable, secure job	EUR 40,000 average, volatile, self-employed
Obligations	None material; owns flat outright	EUR 10,000/yr to ex-wife and children; EUR 100,000 mortgage
Assets	EUR 1,000,000 savings	EUR 10,000 savings
Horizon	20 years to retirement at 50	2-6 years (children aged 12-16 to university)
Questionnaire	Risk tolerant	Risk tolerant
ABILITY	HIGH	LOW
WILLINGNESS	HIGH	HIGH
Conclusion	High risk tolerance overall	Conflict - adopt the lower; invest relatively cautiously

The Jacques case is the instructive one. His questionnaire answers and market views suggest an aggressive investor, but his finely balanced finances and short horizon say otherwise. The adviser's job is to explain the position honestly and then recommend a relatively cautious portfolio anyway.

4.4 Return objectives

Return objectives may be absolute or relative, nominal or real, gross or net of fees, and pre- or post-tax. Whichever is chosen must be stated explicitly and agreed - a nominal objective is easier to measure, a real objective usually relates better to what the client actually wants.

Peer-group objectives - 'top quartile among private equity managers' - are problematic. You rarely know peers' strategies or return methodology, not all investors can be above average, and a good benchmark should be INVESTABLE, which a peer group is not.

Above all the objective must be realistic and consistent with the risk objective. A 15% nominal return may be achievable when inflation is 10%; it is unlikely when inflation is 3%. Where expectations are unrealistic, the adviser must counsel the client on what is actually achievable.

Worked example - Marie Gascon's required return

Step 1 - inflate the target to money of the day:

$$\text{EUR } 2,000,000 \times (1.02)^{20} = \text{EUR } 2,971,895$$

Step 2 - solve for the required growth rate:

$$\text{EUR } 2,971,895 / \text{EUR } 900,000 = (1 + X)^{20}$$

$$X \approx 6.2\% \text{ per year}$$

The EUR 100,000 emergency reserve is EXCLUDED from the calculation - only the EUR 900,000 retirement

portfolio has to reach the target. Watch for this in the exam: money set aside for liquidity is not part of the growth problem.

4.5 The five constraints

1. **LIQUIDITY.** Expected withdrawals - school fees, healthcare, endowment spending rules, insurance claims, pension payments. Cover them with liquid, low-risk assets whose value is known with reasonable certainty at the time the money is needed. Progressive Corporation, an auto insurer facing unpredictable claim timing, holds about 77% in fixed maturities plus 10% short-term investments.
2. **TIME HORIZON.** The period until assets are drawn on or until circumstances change materially. A 55-year-old retiring at 65 has a 10-year horizon even though the portfolio is not liquidated then - its structure must change as income begins. Long horizons support illiquid and risky assets; short horizons do not, because there is no time to recover.
3. **TAX.** Some investors pay tax, others do not. Income is usually taxed more heavily than gains, and gains are typically taxed only on realisation, which gives a time-value benefit. Taxable investors may favour capital gains and, in the US, municipal bonds. Tax-exempt investors such as pension funds are indifferent between income and gains.
4. **LEGAL AND REGULATORY.** Pension allocation limits vary widely - Switzerland caps listed equity at 50% and real estate at 30%; Japan permits 100% equity but prohibits real estate; South Africa allows 75% equity and 25% foreign currency. Self-investment limits cap holdings of the sponsor's own securities. If a client is a director subject to closed-period trading restrictions, the IPS must note it so the manager does not inadvertently trade that stock.
5. **UNIQUE CIRCUMSTANCES AND ESG.** Faith-based restrictions - Shari'a prohibits gambling and interest, so casinos and conventional bonds are excluded. Personal objections to weapons, tobacco, gambling or particular labour and environmental practices. And critically, risks the client already carries outside the portfolio.

KEY IDEA

The unique-circumstances constraint is where the enterprise view enters personal investing. An oil executive should not hold significant oil stocks - his human capital is already an oil exposure. A stockbroker should underweight equities, because his earning power falls exactly when equities do. An entrepreneur should avoid competitors and businesses sharing her risk exposures. This is the Enron lesson written into policy.

Six ESG implementation approaches

Approach	What it does
Negative screening	Excludes companies or sectors on business activity or ESG concerns
Positive / best-in-class	Includes companies with the best ESG performance relative to peers
ESG integration	Systematically considers material ESG factors in allocation, selection and construction
Thematic investing	Invests in themes or assets tied to ESG factors
Engagement / active ownership	Uses shareholder power - voting, proposals, dialogue - to change corporate behaviour
Impact investing	Invests with the intention of generating measurable social or environmental impact alongside a financial return

Screening narrows the investable universe, which changes expected risk and return, so an off-the-shelf benchmark may no longer be appropriate - screened index variants exist to address this. ESG integration is different in kind: because it is a process enhancement rather than a restriction, asset owners increasingly expect managers to beat the STANDARD benchmark while applying it.

The Mountain Materials case shows integration changing a decision. Carbon regulation costs 1% of operating margin, cleaner fuel another 2%, the dividend is cut from 2% to 0%, and weak safety performance implies multiple contraction. The base-case IRR falls from 15% to 5%, and the analyst cuts the position from 1.5% to 0.5% while engaging with management. ESG is one input among many, not an automatic veto.

4.6 Building the strategic asset allocation

IPS objectives and constraints + capital market expectations
→ optimisation / simulation → Strategic Asset Allocation

Capital market expectations are the risk, return and correlation forecasts for each asset class. They are kept separate from the IPS because they require different analysis, different sources and different review cycles.

Two principles justify the focus on asset allocation. First, systematic risk accounts for most of a portfolio's change in value over the long run. Second, returns to groups of similar assets predictably reflect exposure to particular sets of systematic factors. The SAA is therefore the mechanism for buying exposure to systematic risks in the proportions the client needs.

What makes a good asset class

- Assets within it are relatively homogeneous, with similar risk-return expectations
- Paired correlations are HIGH within the class and LOWER against other classes
- Classes are mutually exclusive
- Together they approximate the whole investable universe, so no opportunity is overlooked

Correlation data over December 2000 to August 2018 shows how hard this is for equities: US against European 0.88, US against US small-cap 0.89, US against emerging markets 0.78. Japan is the least correlated at 0.59. Against US Treasuries, however, US equities run at minus 0.37, European at minus 0.28, emerging markets at minus 0.24.

KEY IDEA

Bonds - not another flavour of equity - are the real diversifier. So why do investors still split equities into so many classes? Partly regulation, but mainly organisation: managers specialise by region, size and style, so aligning asset class definitions with available products simplifies implementation.

The optimisation

$$U_p = E(R_p) - \lambda \sigma_p^2$$

$$E(R_p) = \sum w_i E(R_i)$$

$$\sigma_p = \sqrt{(\sum \sum w_i w_j \text{Cov}(R_i, R_j))}$$

$$\text{Cov}(R_i, R_j) = \rho_{i,j} \sigma_i \sigma_j$$

λ measures risk aversion. The optimal allocation is where the efficient frontier just touches the highest attainable indifference curve.

If expectations improve, the frontier shifts up and the tangency point moves, so the allocation should change. If the client's objectives or constraints change, the indifference curves change shape and the tangency point moves again. Neither the frontier nor the client is static.

The Gottschalk case shows how it plays out in practice. A former homebuilder excludes real estate (his income already depends on it) and tobacco (family values). The tobacco screen costs 0.2% of expected European equity return and adds 0.1% of volatility. His objectives are 5% return at 10% risk. No portfolio delivers both exactly - the best available at 10% risk returns 4.9% - and adviser and client agree the shortfall is acceptable. The allocation is 16% European equity, 38% emerging market equity, 46% government bonds. Real plans involve compromise.

4.7 Risk budgeting, drift and rebalancing

Total portfolio risk comes from three sources, and each should have its own limits and expected payoff:

Source	What it is	Rewarded long-run?
Strategic asset allocation	Chosen exposure to systematic risk factors	YES
Tactical asset allocation	Deliberate short-term deviation from policy weights	Only if skilled
Security selection	Picking securities to beat the class benchmark	ZERO-SUM

Security selection is a zero-sum game before costs. All investors in an asset class collectively earn the market return, so one investor's gain is another's loss. After trading costs and salaries, the AVERAGE active manager must underperform. This does not mean skilled managers do not exist, nor that passive managers automatically match their index - a high-turnover index is expensive to track.

Where skill is likeliest to pay is in less efficient markets. US large-cap equity is highly efficient and demands exceptional skill; some regional markets lack the technical and regulatory infrastructure for timely information dissemination, leaving inefficiencies to exploit. Sometimes the choice is made implicitly - non-listed real estate and infrastructure are so illiquid that you cannot buy diversified exposure at all, so participation requires selection.

Drift and the rebalancing policy

As markets move, weights drift from their targets. The rebalancing policy is the set of rules for restoring them - typically a corridor around each policy weight.

Worked example - a European charity

Policy weights: 30% European equity, 15% international equity, 20% government bonds, 20% corporate bonds, 15% cash. Corridor plus or minus 2%.

Asset class	Policy	H1 return	H1 end weight	H2 return
European equities	30%	+15.0%	32.4% BREACH	−9.0%
International equities	15%	+10.0%	15.5%	−6.0%
Government bonds	20%	+0.5%	18.9%	+4.0%
Corporate bonds	20%	+1.5%	19.1%	+4.0%
Cash	15%	+1.0%	14.2%	+2.0%
TOTAL	100%	+6.6%	100%	−2.0%

After six months European equity had breached its 32% ceiling. The committee chose not to rebalance fully back to policy, believing the rally would continue, and merely trimmed to the 32% limit. It did not continue - equities fell while bonds recovered.

$$\text{TAA contribution} = \sum (\text{actual weight} - \text{policy weight}) \times \text{period return}$$

$$\begin{aligned}
 \text{European equity:} & (32.0\% - 30.0\%) \times (-9.0\%) = -0.18\% \\
 \text{International:} & (15.5\% - 15.0\%) \times (-6.0\%) = -0.03\% \\
 \text{Government bonds:} & (18.9\% - 20.0\%) \times (+4.0\%) = -0.05\% \\
 \text{Corporate bonds:} & (19.1\% - 20.0\%) \times (+4.0\%) = -0.04\% \\
 \text{Cash:} & (14.6\% - 15.0\%) \times (+2.0\%) = -0.01\% \\
 \text{TOTAL} & = -0.30\%
 \end{aligned}$$

The tactical decision to stay overweight equities cost 30 basis points. This calculation - actual minus policy weight, times the period return - is a standard exam question.

Two newer developments

- ETFs combined with robo-advice have made cheap, liquid, well-diversified portfolios available to retail investors at a fraction of former costs.
- Risk parity weights asset classes by RISK contribution rather than capital, on the argument that a conventional 60% equity allocation contributes far more than 60% of the risk. Critics note its strong post-2009 record coincided with a long decline in interest rates that flattered bonds.

LEARNING MODULE 5

The Behavioral Biases of Individuals

Traditional theory assumes people consider all available information and act rationally. They do not. Module 5 catalogues how they actually behave and what to do about it.

5.1 The two families of bias

	COGNITIVE ERRORS	EMOTIONAL BIASES
Source	Faulty reasoning: statistical, information-processing or memory errors	Impulse, intuition and feelings
Arise	Through flawed conscious thinking	Spontaneously; may even be unwanted by the person feeling them
Response	MODERATE - reduce or eliminate through better information, education and process	ADAPT - recognise, accept and design around them
Why	Better data and discipline can fix reasoning	You cannot argue somebody out of a feeling

KEY IDEA

Cognitive errors can be corrected; emotional biases can usually only be accommodated. To MODERATE a bias is to reduce or eliminate it. To ADAPT to a bias is to accept it and adjust the plan around it. Which verb applies to which family is examined directly and frequently.

Cognitive errors divide further into BELIEF PERSEVERANCE biases - clinging to what you already think - and PROCESSING ERRORS - using information illogically. Belief perseverance is rooted in COGNITIVE DISSONANCE, the mental discomfort that arises when new information contradicts an existing belief. To relieve it, people ignore or distort the conflicting information.

5.2 Cognitive errors: belief perseverance

Conservatism bias

Maintaining prior views by inadequately incorporating new, conflicting information. In Bayesian terms, overweighting the prior and underweighting the new evidence, so beliefs UNDERREACT.

- Worst where information is technical, abstract or statistical, because the COGNITIVE COST of processing it is higher.
- Fix: apply Bayes' rule deliberately. Ask 'how does this information change my forecast?' Seek help interpreting information you do not understand rather than quietly discounting it.
- Curriculum example: through the 1990s analysts took years to lower Japanese GDP forecasts despite growth having decelerated sharply from 1990.

Confirmation bias

Noticing what confirms existing beliefs and ignoring or undervaluing what contradicts them - a predisposition to justify what we want to believe.

- Consequences: seeing only positive news about a holding; building screens that ignore evidence against their own validity; UNDER-DIVERSIFIED portfolios; and over-concentration in employer stock, because acknowledging bad news would make going to work uncomfortable.
- Fix: actively seek the opposing case and corroborate from a different source or method.
- Example: a portfolio manager whose stock is collapsing calls the analyst who recommended it - who still rates it a buy - rather than an analyst with a sell rating, or researching the competitor taking its market share.

Representativeness bias

Classifying new information by what it resembles. Two forms:

- BASE-RATE NEGLECT - ignoring how often something happens in the wider population in favour of vivid specific information. In the APM case, very few auto parts makers failed in 50 years even in hard times, but alarming headlines lead the analyst to predict failure anyway.
- SAMPLE-SIZE NEGLECT - treating a small sample as representative of the whole population, also called neglecting the law of small numbers.
- Fix: ask explicitly 'what is the probability this belongs to the group it RESEMBLES versus the group it statistically BELONGS to?' Research the base rate; widen the sample.

Illusion of control bias

Believing you can control or influence outcomes you cannot. The classic tell is preferring to choose your own lottery numbers.

- Consequences: concentrated positions in companies people feel they influence, especially employers; OVERTRADING, and research shows portfolio turnover is negatively correlated with returns; and excessively detailed forecasting models, as though more detail controlled uncertainty. Understanding is useful; model complexity does not reduce inherent risk.
- Fix: accept that investing is a PROBABILISTIC activity. Ask what the downside is, what could go wrong, and when you would sell. Speak to someone with an opposing view.

Hindsight bias

Remembering past events as having been predictable, and remembering your own forecasts as more accurate than they were - because outcomes that happened are more available than those that did not. Badly reasoned decisions with good outcomes get remembered as brilliance; well-reasoned decisions with bad outcomes get remembered as avoidable errors.

KEY IDEA

The fix for hindsight bias is mechanical and it works: WRITE DOWN each investment decision and the reasons for it, at the time you make it. Consult the record rather than your memory. Memory will reliably rewrite history in your favour.

5.3 Cognitive errors: processing errors

Anchoring and adjustment

Relying on an initial number and then adjusting insufficiently from it. Closely related to conservatism. The Industrial Lift case: last year's EPS was GBP 1.00 in strong conditions; non-residential construction has since collapsed and a recession is feared; the analyst forecasts GBP 0.90 - a 10% trim from an anchor that conditions have already invalidated.

Fix: ask whether you are holding a position on rational analysis or to reach a price you are anchored to, such as your purchase price or a previous high. Ask whether a forecast is built from past observed quantities or from expected future conditions. Remember that a company's past earnings reflect PAST conditions, and that your own cost basis is irrelevant to the security's future.

Mental accounting

Dividing money into separate mental accounts that influence decisions, even though money is FUNGIBLE. Statman's observation is that investors build portfolios as a layered pyramid, each layer serving a different goal.

- Consequences: missing chances to reduce risk by combining low-correlation assets, because each account is examined separately; irrationally separating income from capital gains and chasing yield while unwittingly risking principal; and treating gains as 'house money' that can be gambled more freely.
- Example: Kendra Liu sells half a position that doubled overnight and puts the proceeds into a high-risk biotech she would never otherwise buy, because it is money she 'did not expect to have anyway'.
- Fix: aggregate. Put every account onto ONE spreadsheet with no headings or labels and look at the true overall allocation. The result is often surprising.

Framing bias

Answering the same question differently depending on how it is presented. One in four start-ups succeeds; three in four fail. Same fact, different decisions.

- GAIN framing tends to produce more risk-averse choices; LOSS framing produces more risk-seeking ones.
- NARROW FRAMING is judging on one or two points - a charismatic CEO - while losing sight of industry conditions, fundamentals and valuation.
- The uncomfortable implication: your own risk-tolerance questionnaire can bias the answer. Showing a 95% return RANGE versus showing a standard deviation, or reordering the risk and return columns, changes which portfolio clients pick.
- Fix: ask whether the decision rests on a net gain or net loss position. Strip out references to gains and losses already incurred and focus on future prospects.

Availability bias

Estimating probability by how easily something comes to mind. Four sources:

- RETRIEVABILITY - the first answer to surface is taken as correct.
- CATEGORIZATION - searching only within familiar categories.
- NARROW RANGE OF EXPERIENCE - generalising from your own limited exposure. The hedge fund employee who concludes 'most CFA charterholders work at hedge funds'.

- **RESONANCE** - being swayed by how closely something mirrors your own situation.

Consequences: a limited opportunity set; choosing funds by advertising volume or news coverage; and failing to diversify away from your own industry. Fix: build a proper policy strategy, research before deciding, use long-term historical data, and ask directly how the candidate list was assembled.

5.4 Emotional biases

Loss aversion and the disposition effect

Losses hurt considerably more than equivalent gains please. The value function is S-shaped and asymmetric around a reference point, which produces **RISK-SEEKING** behaviour in the domain of losses and **RISK-AVOIDANCE** in the domain of gains - the opposite of rational behaviour, which would accept more risk to increase gains rather than to avoid losses.

KEY IDEA

The **DISPOSITION EFFECT**: holding losers too long and selling winners too soon. The consequence is not merely poor returns - the portfolio becomes **RISKIER** than the client's objectives justify, because the deteriorating positions are kept and the improving ones are sold.

Overconfidence bias

Unwarranted faith in one's own abilities. Two forms:

- **PREDICTION** overconfidence - confidence intervals set far too narrow, with too little variation allowed around a forecast.
- **CERTAINTY** overconfidence - probabilities assigned to outcomes set too high. The analyst who says 'there is no credible downside case' when asked what happens if oil falls another 10%.

Amplified by **SELF-ATTRIBUTION BIAS**: taking credit for successes (self-enhancing) while blaming others for failures (self-protecting). Consequences are underestimated risk, overestimated return, and poorly diversified portfolios with significant downside.

Fix: review your actual trading record over at least two years, winners **AND** losers, and count the trades. Separate decisions that were genuinely good from those that were merely lucky. Look for repeated patterns in the losses. As the old Wall Street line has it - do not confuse brains with a bull market.

Self-control bias

Failing to act on long-term goals because of short-term satisfaction, worsened by **HYPERBOLIC DISCOUNTING** - a strong preference for small payoffs now over larger payoffs later. Consequences: saving too little, then taking excessive portfolio risk to make up the shortfall; and borrowing too much to fund present consumption. Fix: a written plan and a budget, reviewed regularly, with a strategic asset allocation to anchor decisions.

Status quo bias

Doing nothing when change is warranted - through INERTIA rather than any conscious choice. The distinction matters: endowment and regret-aversion biases also produce inaction, but through conscious, if mistaken, decisions.

The Choi et al. study of automatic 401(k) enrolment is the evidence. Enrolment jumped from 26-43% at six months and 57-69% at three years to over 85% at all three firms. But more than 65% of employees simply contributed the employer's default of 2% or 3% and stayed in the default investment option, and after two years over 40% were still on the default. Defaults are extraordinarily powerful.

Endowment bias

Valuing an asset more simply because you own it, so that the minimum price you would sell at exceeds the maximum you would pay to buy the same thing. It applies to purchased as well as inherited holdings.

KEY IDEA

The single most effective question in Module 5: 'Would you buy this security TODAY at the current price?' If the honest answer is no, endowment bias is driving the holding. For inherited assets, ask whether the deceased's intent was to leave that specific portfolio, or simply to leave resources that benefit the heirs. Most people conclude the latter, and become open to changing the allocation.

Regret-aversion bias

Avoiding decisions for fear they turn out badly. Regret is more intense for actions TAKEN than for actions NOT taken, so inaction becomes the default.

- Being too conservative after past losses, leading to long-run underperformance and missed goals.
- HERDING - crowding into popular, well-known names because unfamiliar choices feel riskier and carry more personal responsibility. Keynes: 'worldly wisdom teaches that it is better for reputation to fail conventionally than to succeed unconventionally'.
- Fix: quantify the benefits of diversification and proper allocation. Accept that losses happen to everyone; keep long-term objectives in view to avoid being either too cautious or swept into a bubble.

5.5 Biases at the market level: anomalies

An ANOMALY is a persistent abnormal return that is predictable in direction. But not every apparent deviation is genuine. Three sources of false positives:

1. The asset pricing model chosen. If a reasonable change in how you estimate normal returns makes the anomaly vanish, it was an illusion - Fama places the IPO puzzle and post-stock-split returns in this category. Persistent high returns may simply be compensation for risk.
2. Statistical problems - small samples, selection or survivorship bias, and data mining that treats spurious correlations as real. Benchmark choice alone can change the size of any measured out- or underperformance.

3. Temporary disequilibrium. Publication draws attention and arbitrage removes the pattern. The small-company January effect does not survive risk adjustment, and the weekend effect has faded in the US and UK.

Momentum

Future prices correlate with the recent past, typically for up to two years before reversing. The London Business School study of 52 years of UK data found the best performers of the prior 12 months went on to return 18.3% annualised while the worst returned 6.8%, against a market return of 13.5%. The effect appeared in all 16 other international markets studied.

Behavioural explanations: AVAILABILITY in the form of the RECENCY EFFECT - recent events are recalled vividly and extrapolated, with private investors more prone to this while professionals more often expect mean reversion. And REGRET, an expression of HINDSIGHT BIAS - investors who feel they should have foreseen a move buy in to remedy the regret, which also contributes to overtrading.

Bubbles and crashes

Some elements are rational. Investors may expect a crash without knowing its timing; shorting may be costly, capital may be unwilling to bear extended losses, or suitable instruments may not exist. And managers judged on short-term performance face real commercial and career risk for standing aside.

The curriculum's paired example is the sharpest illustration in the volume. The manager of Hedge Fund A knew technology stocks were overvalued by December 1999 but misjudged the timing - 'we thought it was the eighth inning, and it was the ninth' - and resigned in April 2000 despite a strong 12-year record. The manager of Hedge Fund B refused to buy technology stocks in 1998 and 1999 because he correctly judged them overvalued, and after 17 years of strong performance the fund was DISSOLVED in 2000 because its returns could not keep up.

EXAM TRAP

Being right is not sufficient. Fund B's manager made the correct call and lost his fund for it. This is why career risk is treated as a RATIONAL explanation for bubble participation, not merely a behavioural one.

The biases at work during bubbles: OVERCONFIDENCE above all - overtrading, underestimating risk, failing to diversify, rejecting contradictory information. In a rising market almost every sale is profitable, so CONFIRMATION and SELF-ATTRIBUTION biases convert luck into perceived skill. REGRET AVERSION draws in those who feel they are missing out. As the bubble unwinds, ANCHORING makes markets underreact, cognitive dissonance leads investors to rationalise losses, and eventual capitulation accelerates the decline.

Value versus growth

Fama and French found value stocks - high book-to-market, low P/E, low price-to-dividend - beat growth stocks in 12 of 13 major markets over 1975-1995. But the anomaly DISAPPEARS in a three-factor model, suggesting size and book-to-market are compensation for risk, such as greater vulnerability to distress in downturns.

The behavioural counter-explanation is the HALO EFFECT, a form of representativeness: a favourable evaluation of one characteristic spreads to others, so a company with a good growth

record and strong past share performance is assumed to be a good investment with higher expected returns than its risk warrants. Overconfidence in growth forecasts compounds it.

Related is HOME BIAS - a preference for domestic securities, and even for companies headquartered near the investor. Explanations include a perceived informational advantage, comfort from access to management, and a desire to invest locally. Note also the perverse finding that a more positive emotional rating leads investors to perceive a stock as LESS risky: CAPM says risk and expected return move together, but many investors behave as though they move in opposite directions.

LEARNING MODULE 6

Introduction to Risk Management

Risk management is not risk minimisation. It is choosing risks deliberately, sizing them, measuring them continuously, and keeping them aligned with a tolerance decided in advance.

6.1 What risk management actually is

Risk management is the process by which an organisation or individual defines the level of risk to be taken, measures the level of risk being taken, and adjusts the latter toward the former - with the goal of maximising the entity's value or the individual's utility.

Note the three verbs: DEFINE, MEASURE, ADJUST. Nothing in the definition mentions reducing or avoiding risk.

KEY IDEA

A company that shied away from all risk would find it could not operate. Risk management is not about avoiding risk any more than a practical diet is about avoiding calories.

Three meanings of the word 'risk'

The curriculum separates them carefully, using a yen example. An announcement will move the yen up or down by 1%, and you hold JPY 1,000,000.

Term	In this example	Meaning
Risk driver	The uncertain $\pm 1\%$ move	The underlying uncertainty
Risk position	JPY 1,000,000 held	The size of the risky action taken
Risk exposure	$\pm$ JPY 10,000	The potential valuation change

Here exposure is simply position times driver. In practice all three get called 'risk', which is why questions often turn on which one is meant.

The Doctrine of No Surprises

A risk manager is not expected to predict the crisis. They are expected to have already quantified what the crisis would cost. Before 2008 a good bank risk manager would not have known a real estate crisis was coming, but would have told the board that such a crisis could destroy 60% of capital, forced a governance discussion about whether that was tolerable, and hedged the portion that was not. The only surprise should be the market shock itself.

KEY IDEA

In a good risk management process most of the work happens BEFORE the event. In a poor one just as much work gets done - but all of it afterwards, once the damage is already done.

The benefits of doing it well: fewer surprises and a known cost when they occur; better decision discipline and risk-return trade-offs; faster response through active monitoring; fewer operational errors; more trust between the board and management, which supports effective delegation; and a better reputation with analysts and investors. Together these raise enterprise value.

6.2 The risk management framework

1. **RISK GOVERNANCE** - the top-down foundation. Board-level. Sets goals, defines risk tolerance, provides oversight.
2. **RISK IDENTIFICATION AND MEASUREMENT** - the analytical core. Analysing risk drivers, tracking exposures, computing metrics under scenarios and stresses. Includes qualitative assessment, not only quantitative.
3. **RISK INFRASTRUCTURE** - the people and systems. Risk capture, databases, models, a stress engine, reporting, and skilled staff. With heavier reliance on technology, more testing is needed to avoid the irony of errors originating inside the risk system itself.
4. **POLICIES AND PROCESSES** - governance extended into daily operations. Limits, constraints, due diligence, escalation procedures, checklists.
5. **RISK MONITORING, MITIGATION AND MANAGEMENT** - the day-to-day work. Recognising when exposure has drifted outside tolerance and acting.
6. **COMMUNICATION** - continual, across all levels, with a feedback loop back to the governing body. Reporting matters MORE, not less, when limits are breached.
7. **STRATEGIC ANALYSIS AND INTEGRATION** - the offensive use. Sorting out which activities add value and which do not, and feeding that back into decisions.

For an individual the same framework applies in reduced form: set goals; choose investments and identify their risks; evaluate exposure; consider ways to modify it; implement; and review periodically. The individual acts as their own governing body while an adviser performs the management role. The curriculum's warning is that individuals often decide risk management is too complicated and skip it - at essentially the same cost as a corporation would face, only with less money involved.

When risk is genuinely integrated into every decision rather than bolted on afterwards, the organisation has an effective **RISK CULTURE**.

6.3 Risk governance and risk tolerance

ENTERPRISE RISK MANAGEMENT means focusing risk activity on the objectives, health and value of the **WHOLE** organisation, using the entire economic balance sheet rather than assets alone.

Why the enterprise view matters - the pension case

A pension fund manager maximising only the fund's ASSETS may ruin the sponsor. Pension liabilities are bond-like, so an all-equity fund faces a market collapse and falling interest rates simultaneously: assets fall while liabilities rise in value, and the fund becomes insolvent.

A true enterprise view goes further still and considers the PARENT COMPANY. In that same collapse the sponsor is probably in recession, so demands for extra contributions arrive exactly when they are hardest to meet. Factoring in the corporate risk profile lowers the appropriate risk tolerance again.

For individuals the equivalent is total wealth, not just the portfolio. Someone with a career in real estate should hold fewer real estate securities. And three retirees - one with an inflation-linked pension, one with a fixed pension, one with no pension at all - require completely different solutions. Because of the variability of an individual life cycle and the discreteness of personal goals, the curriculum argues the enterprise view is EVEN MORE important for individuals than for institutions.

Governance structures

- A RISK MANAGEMENT COMMITTEE gives senior decision-makers a regular forum, paralleling the board's deliberations at an operational level. It does NOT approve the governing body's policies.
- A CHIEF RISK OFFICER builds and runs the framework and participates in strategic decisions - this is not purely a policing role. The curriculum's argument: it makes no more sense for the CEO to act as CRO than to act as CFO.

Setting risk tolerance

Risk tolerance identifies the extent to which the organisation is willing to experience losses or opportunity costs and to fail in meeting its objectives. It requires two analyses combined:

- The INSIDE view - what shortfalls would cause us to fail, or to miss critical goals?
- The OUTSIDE view - what uncertain forces are we exposed to? What are our risk drivers?

The Spanish construction equipment example makes it concrete. Inside: a 5-10% revenue drop is survivable, but a 20% drop triggers debt covenants and imperils a flagship product launch; EUR 40 million of annual cash flow is needed for critical capital expenditure and almost none of it can be at risk. Outside: three uncontrollable drivers - the US dollar, interest rates, and industrial sector equity returns. The board therefore caps cash flow variation at EUR 10 million a year and revenue exposure at minus 10% in a global recession, which then shapes financing choices and drives a hedging programme.

EXAM TRAP

Risk tolerance must be decided and communicated BEFORE a crisis. Many organisations only hold the discussion afterwards, which the curriculum compares to buying insurance after the loss has occurred.

Factors that SHOULD drive risk tolerance	Factors that often do but SHOULD NOT
Goals and strategy	Personal motivations and beliefs of board members (the agency problem)
Areas of genuine expertise	Company size
Ability to respond dynamically to adverse events	Whether markets currently seem stable
Loss the firm can sustain as a going concern	Short-term pressures
Competitive and regulatory landscape	Management compensation arrangements

Good governance also steers **WHERE** risk is taken. Pursue risk in areas of core competence, where the firm is positioned to create value; limit or hedge non-core risks where it has no comparative advantage. Companies that take risk in areas where they have no expertise put their core value creation - and sometimes the whole organisation - at peril.

It is easy to find strategies producing outsized short-term returns at extreme risk. Selling put options on your own equity raises short-term profits and dramatically increases the chance of failure in a market decline; excessive leverage does the same. A formal risk tolerance naturally steers the discussion away from strategies that simply trade ruin for return.

6.4 Risk budgeting

Where risk tolerance says **HOW MUCH**, risk budgeting says **HOW AND WHERE**. It quantifies and allocates the tolerable risk using specific metrics, bridging the gap between one high-level board decision and thousands of operating choices.

The key shift is to view a portfolio by risk characteristics rather than by product labels. A portfolio described as 20% hedge funds, 30% private equity and 50% stocks and bonds might equally be described as 70% driven by global equity returns, 20% by domestic equity and 10% by interest rates, with 45% illiquid. The two descriptions coexist, and the risk view is usually more informative - some equities are low risk and some hedge funds are very high risk, so the product label tells you little.

- Single-dimension budgets: standard deviation, beta, value at risk, or scenario loss. Even the simplest measure delivers real benefits.
- Multi-dimensional budgets: allocation by underlying risk class, or by factor exposures - a beta target with value and momentum tilts layered on top.

KEY IDEA

The hidden benefit of risk budgeting is that it forces trade-offs into every decision. Once the budget binds, you must choose where return per unit of risk is highest, and you must compare each active bet against simply buying the market and hedging. Every risky decision is measured against a passive alternative on a risk-equivalent basis - so the framework pushes you to add value in each decision rather than merely to be paid for risk.

Individuals frequently fail here without noticing. The classic case is holding employer stock in a personal portfolio, which concentrates the total wealth budget - financial plus human capital - into one firm and one industry. It almost never comes from a formal plan; it comes from

inaction.

6.5 Identifying risks

Financial risks - arising from the financial markets

Risk	Definition and notes
Market risk	Movements in interest rates, stock prices, exchange rates and commodity prices. The most visible risk, with abundant data - risk management knowledge is most advanced here.
Credit risk	Loss if a counterparty fails to pay an obligation. Also called default or counterparty risk. With swaps and forwards either party can end up owing, so credit risk is BILATERAL. Defaults have far longer-lasting consequences than price falls, which can reverse.
Liquidity risk	A significant downward valuation adjustment when selling. Note carefully: a known bid-ask spread is a COST, not a risk. The RISK is the UNCERTAINTY of that spread - it could be called transaction cost risk.

Liquidity risk has two sources: market liquidity varies over time, and the cost and uncertainty of liquidation rise with position size relative to normal trading volume. In extreme cases there may be no price above zero at which the asset can be sold.

Non-financial risks

- **SETTLEMENT RISK** - you pay, the counterparty goes bankrupt before delivering. Also called Herstatt risk, after the German bank that failed in 1974 after receiving overnight payments. Often arises from time zone differences.
- **LEGAL RISK** - being sued, or a contract not being upheld. Even a seemingly weak case can prevail in court.
- **REGULATORY, ACCOUNTING and TAX RISK** - collectively compliance risk. Rules always lag financial innovation, so updates bring unexpected costs, back taxes, restatements and penalties.
- **MODEL RISK** - valuation error from using the wrong model, or the right model wrongly. Assuming constant dividend growth when growth is not constant is the curriculum's simplest example.
- **TAIL RISK** - more extreme events than probability models predict. Ignoring tail risk is itself a form of model risk, arising from internal modelling choices.
- **OPERATIONAL RISK** - inadequate or failed people, systems and internal processes, plus external events affecting operations.
- **SOLVENCY RISK** - running out of cash, even while otherwise solvent.

How badly the normal distribution fails

S&P 500 monthly returns, January 1950 to October 2018: mean 0.70%, standard deviation 4.10%. Under a normal distribution:

Month	Return	Expected frequency under normality
October 1987	-21.76%	Once every 2,199,935 years
October 2008	-16.94%	Once every 6,916 years
August 1998	-14.58%	Once every 654 years
October 1974	+16.30%	Once every 888 years

EXAM TRAP

All three of those crashes occurred inside a single 68-year window. If the normal distribution described returns, they should never have happened. It is safe to reject normality for at least another two million years - yet option models, portfolio construction and asset allocation routinely assume it. This is why market risk is so often dealt with in an oversimplified way.

Operational risk in detail

- Employees - theft, but also honest mistakes. Crediting USD 100,000 for a USD 100 deposit is an innocent error that can lose money very fast. Accounting fraud has often been committed to make a company look better rather than for personal gain.
- Rogue traders - personified by Nick Leeson, whose speculative trades destroyed the 200-year-old Barings Bank in 1995. The defining feature is trading without regard for the organisation's limits or controls.
- Business interruption - floods, earthquakes, hurricanes. External and uncontrollable, but that does NOT excuse failing to prepare. Generators, backup facilities and remote working are cheap.
- Cyber risk - hackers succeed only where systems are vulnerable, so security is the organisation's responsibility. Under GDPR, breaches of sensitive personal data must be notified within 72 hours, with fines of several million euros for failure - and this applies to organisations outside the EU that target European citizens.
- Terrorism - after the 1993 World Trade Center attacks many firms established distant backup facilities, which proved decisive in 2001.

Solvency risk

The most underappreciated risk before 2008. Lehman Brothers is usually described as a leverage failure, and leverage certainly contributed, but what actually forced the bankruptcy was solvency: funding disappeared almost overnight once counterparties refused Lehman's credit risk. Even a day of large market GAINS would not have saved it - it had already been destroyed by solvency risk.

Solvency risk is the clearest argument for the enterprise view. A university endowment might hold a perfectly balanced portfolio in isolation, but in a deep recession tuition revenue, grants and donations all fall at the same time as portfolio values and distributions - forcing emergency actions that impair the endowment purely because the wider enterprise needs cash.

It is easily mitigated but never eliminated, and none of the safeguards is free: less leverage, more stable financing, better transparency, holding more cash equivalents and fewer illiquid

assets.

Risks specific to individuals

- **HEALTH RISK** - direct costs, reduced income through disability, reduced lifespan or quality of life.
- **MORTALITY RISK** - dying young, ending the income stream a family depends on.
- **LONGEVITY RISK** - outliving your resources. Insurers and DB plans only need reliable GROUP averages from mortality tables; an individual has no idea which side of the average they will fall on, which is why DC investors face a genuinely harder problem.
- **PROPERTY AND CASUALTY** - fire, natural disaster, liability from harming others.

6.6 Risks interact - and they compound

KEY IDEA

Market risk begets credit risk, which begets operational risk. Risks are almost never independent, and the combined risk is practically always worse than the sum of the parts. Most risk models do not capture the interaction, which makes the consequence worse still.

Wrong-way risk

You buy an out-of-the-money put with a JPY 1,000 strike from Counterparty C. Assuming a 2% default probability INDEPENDENT of the market, the contract prices at about JPY 98 rather than JPY 100.

But C's default probability is not independent. It is far higher precisely when the equity market has collapsed - which is exactly when the put pays. The real default probability in that state might be 10% or more. In the extreme event where you should receive JPY 1,000, you will very likely receive nothing. You bore far more risk than you thought and you OVERPAID for the contract.

This pattern was widespread in 2008, when holders of mortgage credit securities believed their risks were well diversified when in truth they were systematic.

Non-linearity

Baseline loss	Outcome for a 2× levered firm with liquidity strain
1%	2% - simple leverage, as expected
10%	25% rather than 20% - liquidity and funding stress begins to bite
30%	Failure - the toxic interplay of leverage and liquidity

This is what happened to many banks and funds in 2008 and to Long-Term Capital Management in 1998: leverage manifesting as higher market risk, interacting toxically with liquidity and solvency risk. It is also why up-front SCENARIO PLANNING is so valuable - linear models simply do not capture it.

The individual equivalent is employer stock again: market risk and human capital risk interacting so that a single event takes the job and the savings together.

6.7 Measuring risk

Risk drivers originate in the global and domestic macroeconomy, in industries, and in individual companies. Risk management can influence some of this but not all of it - a risk manager can reduce the chance their own firm defaults, but cannot control interest rates. For the latter, the job is to position the firm so its exposure matches its tolerance.

Measure	What it captures
Probability	Relative frequency of an outcome. Necessary but far from sufficient on its own.
Standard deviation	Dispersion. Widely used but may not exist for fat-tailed distributions, and overstates risk in a diversified context.
Beta	Systematic risk - how much market risk an asset adds to a diversified portfolio.
Delta	First-order: derivative price sensitivity to a SMALL move in the underlying.
Gamma	Second-order: how delta itself changes. Captures LARGE moves.
Vega	First-order sensitivity to the volatility of the underlying.
Rho	Sensitivity to interest rates. Low for most options, high for interest rate options.
Duration	First-order interest rate sensitivity of a fixed-income instrument - the bond analogue of delta.

Value at risk

"Our VaR is GBP 3 million at 5% for one day."

= we expect to lose AT LEAST GBP 3 million in one day, 5% of the time
= a minimum loss of GBP 3 million about once every 20 business days

Three components: an amount, a time period, a probability.

EXAM TRAP

VaR is a MINIMUM extreme loss, not a maximum. The word 'minimum' is the most frequently overlooked word in the definition and the most frequently tested. There is no maximum in a VaR measure - the true maximum is the entire equity of the organisation.

VaR is simple, accepted by most banking regulators, approved for accounting disclosure - and controversial. Different estimation methods give widely different answers. It is subject to model risk: wrong distributional assumption or wrong inputs, wrong VaR. Critics argue naive users are lulled into false security by a tolerable-looking number while the real risk is uncontrolled. Always supplement it.

- CVaR (conditional VaR) - the weighted average of ALL loss outcomes exceeding the VaR loss. It tells you how bad the tail is, not merely where it begins.

- **EXPECTED LOSS GIVEN DEFAULT** - the credit analogue: if this issuer defaults, how much do we lose on average?
- **EXTREME VALUE THEORY** - statistics devoted to the tails, long used by insurers.
- **SCENARIO ANALYSIS** - a package of related stress tests under a common theme, such as a sharp rate rise combined with a currency collapse.
- **STRESS TESTING** - specific extreme price moves, rare but potentially destabilising for the whole organisation. Now required of major banks by the US Federal Reserve and other central banks.

Measuring credit and operational risk

Credit analysis leans on rating agencies but also on liquidity ratios such as the current ratio, solvency ratios such as interest coverage, profitability measures such as return on assets, and leverage measures such as debt to total assets - plus the strength and cyclicity of the industry.

The structural difficulty is that credit events are RARE for any one issuer. Very few companies that default have a history of defaulting. Lehman Brothers had operated since 1850 without one - imagine assigning it a default probability in 2007. Risk managers therefore pool similar companies, which is essentially what a credit rating does. Market prices of CREDIT DEFAULT SWAPS, options and insurance contracts provide a valuable second signal: the market's own ex ante price of the risk.

Operational risk is harder still. Assessing the probability of an event such as the 2014 Home Depot card breach - and the years of litigation risk that follow - is close to impossible from internal data alone, so third parties aggregate events across many companies and publish statistics. Regulatory, accounting and political risks often admit no numeric measure at all and revert to subjective judgement.

6.8 Modifying risk

Modification is not always reduction. A portfolio targeting 50/50 equity and cash drifts toward cash when cash outperforms, and beyond a point its risk becomes unacceptably LOW given the return target - so rebalancing means INCREASING risk.

Method	What it means	Typical tool
Prevention and avoidance	Do not take the risk at all	Strategy, exclusions, governance limits
Acceptance	Bear it, but efficiently	Self-insurance, reserves, capital, DIVERSIFICATION
Transfer	Pass it to another party	INSURANCE, surety and fidelity bonds, reinsurance, catastrophe bonds
Shifting	Change the shape of the payoff distribution	DERIVATIVES

Prevention and avoidance

Rarely as simple as it sounds, because avoiding risk usually means avoiding the benefit too. You could eliminate car accident risk by never travelling by car; you could protect children from all harm at the cost of preparing them poorly for adult life; you could hold only cash and forgo

inflation protection and long-term growth.

Nearly every risk has an upside as perceived by the risk taker - which is why people gamble, smoke or skydive despite the apparent costs. These are simply cases of choosing to bear risk, conceptually identical to an investor accepting a high degree of portfolio risk. In organisations this is a strategic, board-level decision, often used to exclude areas so management can concentrate on where it can add value.

Acceptance: self-insurance and diversification

Self-insurance means bearing a risk considered undesirable but too costly to remove - sometimes simply bearing it, sometimes establishing a reserve. A young, healthy adult without dependants who declines health insurance and sets money aside is fully self-insuring. Banks self-insure through capital and loan loss reserves.

EXAM TRAP

There is a fine line between self-insurance and DENIAL. Reserving against a loss that sits inside your risk tolerance is good governance. An investment firm with a EUR 1 billion loss tolerance that leaves a EUR 3 billion rogue trader exposure unhedged and calls it 'self-insuring' is simply ignoring the risk and violating its own governance.

Transfer: insurance

Insurance works through POOLING of low-correlation risks. The insurer charges a premium covering expected aggregate losses, operating costs and profit, using actuarial data that is generally readily available. A well-diversified insurer does not care if one policyholder claims more than average, provided claims are uncorrelated.

- Insurers manage their own risk by limiting concentration - not writing too much Gulf Coast hurricane cover - and by REINSURANCE, selling risk to another insurer. A Midwest tornado insurer may happily accept some hurricane risk in exchange.
- CATASTROPHE BONDS pass insurance risk to bond investors, who may lose principal or interest if claims exceed a threshold.
- DEDUCTIBLES serve three purposes: they cut the number of small claims, each of which carries a fixed processing cost; they encourage good risk management by the insured; and they let the insured combine transfer with self-insurance for a better overall trade-off.
- Where risks CANNOT be pooled - one Olympics, one volatile film star - specialist cover through Lloyd's syndicates is required, with investors bearing losses directly. Because such risks are uncorrelated with each other, writing several of them still achieves some diversification.
- SURETY BONDS pay if a third party fails to perform; FIDELITY BONDS cover employee dishonesty. In this context 'bond' means an assurance, not a debt security.

Shifting: derivatives

Risk transfer passes risk to another party; risk SHIFTING changes the distribution of outcomes - adjusting the payoff diagram itself. A company willing to make slightly less profit if markets rise, in exchange for not losing more than 20% if they fall, is shifting risk. This is the bulk of hedging and the most common form of risk modification for financial firms.

	Forward commitments	Contingent claims (options)
Examples	Forwards, futures, swaps	Calls and puts
Obligation	BOTH parties obligated	Buyer has the right, not the obligation
Cash at inception	NONE	Premium paid up front
Outcome	Locks in a price or rate; no flexibility	Keeps the upside, caps the downside
Trade-off	Free, but you give up favourable moves	Flexible, but you pay for it

Worked hedging example

A UK investor holds 60% FTSE 100 and 40% US Treasuries. Unhedged, the portfolio expects 3.9% with 8.9% risk. Hedging 100% of the currency exposure with a one-year forward:

	Unhedged	Currency hedged
Expected return	3.90%	3.87%
Risk (standard deviation)	8.9%	7.6%

Risk falls by 1.3 percentage points for 3 basis points of return - because the one-year forward price of USD in GBP, 0.7038, is 0.03% below the spot of 0.7040, and 40% of the portfolio is hedged.

EXAM TRAP

The hedge is NOT perfect and this detail is examinable. As the Treasury investment changes value it will be worth more or less than the USD amount sold forward, leaving the portfolio over- or under-hedged. Because the investment is expected to GROW, being under-hedged is the likely outcome. Managing it means comparing the forward's notional value against the investment at intervals - and how often to do so is itself a trade-off between monitoring cost, transaction cost and residual risk.

Choosing a method

No method has a clear advantage; they are not mutually exclusive, and most organisations use all four. Some airlines hedge fuel and some do not; among those that do, some prefer the certainty of forwards and swaps and others the flexibility of options. Some manufacturers hedge currency with derivatives while others build plants abroad or match currency assets against liabilities.

1. PREVENT AND AVOID first, for risks offering few benefits against potentially extreme costs - especially those outside core competence. But avoidance may not be good value, and avoiding risk often means avoiding opportunity.
2. SELF-INSURE where you can afford it. It avoids external monitoring costs and gives the greatest flexibility. Few organisations have enough cash to self-insure everything, and some risks can imperil the entire capital base.

3. TRANSFER where risks pool effectively and the premium is less than the expected benefit. Many risks are not insurable cost-effectively, particularly those affecting many parties at once.
4. SHIFT with derivatives for financial risks beyond appetite - the most common choice in that category, though not every risk has a derivative available.

KEY IDEA

The methods leave DIFFERENT RISK PROFILES behind, not merely different amounts of risk. A contingent claim hedge and a forward commitment hedge produce genuinely different shapes of outcome. Ultimately the decision is always balancing costs against benefits while producing a risk profile consistent with the organisation's risk tolerance and governance objectives.

Appendix A - Formula sheet

Everything from Volume 9 you may need to compute under exam conditions.

Module 1 - Portfolio risk and return

Expected return build-up

$$1 + E(R) = (1 + r_{rF}) \times [1 + E(\pi)] \times [1 + E(RP)]$$

Utility

$$U = E(r) - \frac{1}{2} A \sigma^2$$

Two-asset portfolio return

$$R_p = w_1 R_1 + (1 - w_1) R_2$$

Two-asset portfolio risk

$$\sigma_p = \sqrt{(w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2)}$$

Covariance

$$\text{Cov}(R_1, R_2) = \rho_{12} \sigma_1 \sigma_2$$

Three-asset portfolio risk

$$\begin{aligned} \sigma_p = \sqrt{(w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + w_3^2 \sigma_3^2} \\ + 2\rho_{1,2} w_1 w_2 \sigma_1 \sigma_2 + 2\rho_{1,3} w_1 w_3 \sigma_1 \sigma_3 + 2\rho_{2,3} w_2 w_3 \sigma_2 \sigma_3) \end{aligned}$$

Equally weighted N-asset portfolio

$$\sigma_p^2 = \bar{\sigma}^2/N + [(N-1)/N] \times \text{Cov}$$

Capital allocation line

$$E(R_p) = R_f + [(E(R_i) - R_f) / \sigma_i] \times \sigma_p$$

Test for adding a new asset

$$[E(R_{\text{new}}) - R_f] / \sigma_{\text{new}} > \{[E(R_p) - R_f] / \sigma_p\} \times \rho_{\text{new},p}$$

Foreign asset return in domestic currency

$$R_D = (1 + R_{lc}) \times (1 + R_{FX}) - 1$$

Module 2 - CAPM and performance

Capital market line

$$E(R_p) = R_f + [(E(R_m) - R_f) / \sigma_m] \times \sigma_p$$

Kinked CML with a higher borrowing rate R_b

$$w_1 \geq 0: E(R_p) = R_f + [(E(R_m) - R_f) / \sigma_m] \sigma_p$$

$$w_1 < 0: E(R_p) = R_b + [(E(R_m) - R_b) / \sigma_m] \sigma_p$$

Risk decomposition

$$\sigma_i^2 = \beta_i^2 \sigma_m^2 + \sigma_e^2$$

Beta

$$\beta_i = \text{Cov}(R_i, R_m) / \sigma_m^2 = \rho_{i,m} \times (\sigma_i / \sigma_m)$$

Market model

$$R_i = \alpha_i + \beta_i R_m + e_i$$

CAPM / security market line

$$E(R_i) = R_f + \beta_i [E(R_m) - R_f]$$

Portfolio beta

$$\beta_p = \sum w_i \beta_i$$

Sharpe ratio $(R_p - R_f) / \sigma_p$

Treynor ratio $(R_p - R_f) / \beta_p$

M^2 $(R_p - R_f)(\sigma_m / \sigma_p) + R_f$

M^2 alpha $M^2 - R_m$

Jensen's alpha $\alpha_p = R_p - [R_f + \beta_p(R_m - R_f)]$

Security characteristic line

$$R_i - R_f = \alpha_i + \beta_i(R_m - R_f)$$

Non-market security weighting: proportional to α_i / σ_{ei}^2

Information ratio: α_i / σ_{ei}

Modules 3 to 6**Diversification ratio****SD of equally weighted portfolio / SD of a randomly selected security****Utility used in asset allocation**

$$U_p = E(R_p) - \lambda \sigma_p^2$$

Multi-asset expected return

$$E(R_p) = \sum w_i E(R_i)$$

Multi-asset risk

$$\sigma_p = \sqrt{(\sum \sum w_i w_j \text{Cov}(R_i, R_j))}$$

Inflating a return target to money of the day

$$FV = PV \times (1 + \text{inflation})^n$$

Required rate of return

$$FV / PV = (1 + X)^n \rightarrow \text{solve for } X$$

Tactical asset allocation contribution

$$\sum (\text{actual weight} - \text{policy weight}) \times \text{period return}$$

Appendix B - The distinctions most often confused

If you lose marks in Volume 9, it will almost certainly be on one of these.

Confusion	The distinction that resolves it
Risk aversion vs risk tolerance	Opposites. High tolerance = low aversion = low A.
Ability vs willingness to take risk	Ability is objective (horizon, income, wealth vs liabilities). Willingness is psychological. On conflict, adopt the LOWER and document it.
CML vs SML	CML: total risk (σ) on the x-axis, efficient portfolios only. SML: systematic risk (β), ANY security. Equal only for efficient portfolios.
Sharpe / M^2 vs Treynor / alpha	Total risk vs systematic risk. Not diversified $\rightarrow$ Sharpe or M^2 . Well diversified $\rightarrow$ Treynor or Jensen's alpha.
Total risk = systematic + non-systematic	These add as VARIANCES, never as standard deviations.
High standard deviation vs high beta	Unrelated. Gold in the curriculum has market-level SD and a beta of ZERO, because correlation is zero.
Sharpe ratio vs Jensen's alpha	Sharpe is meaningless alone and must be compared. Alpha is interpretable on its own - positive means outperformance.
Cognitive errors vs emotional biases	Cognitive: faulty reasoning, MODERATE them. Emotional: feelings, ADAPT to them.
Status quo vs endowment vs regret aversion	All produce inaction. Status quo is INERTIA; the other two are conscious but mistaken choices.
Conservatism vs anchoring	Conservatism = under-updating a BELIEF. Anchoring = adjusting insufficiently from a starting NUMBER.
Base-rate vs sample-size neglect	Both are representativeness. Base-rate ignores population frequency; sample-size over-trusts a small sample.
Liquidity risk vs solvency risk	Liquidity: uncertainty of the price concession on SELLING an asset. Solvency: running out of CASH to operate. Practitioners often call the second one liquidity risk too.
Credit risk vs settlement risk	Credit: the counterparty cannot pay. Settlement: you paid first and they failed before delivering (Herstatt risk).
VaR minimum vs maximum	VaR is a MINIMUM extreme loss at a given probability. There is no maximum short of total equity.

Confusion	The distinction that resolves it
Risk transfer vs risk shifting	Transfer = insurance, works by POOLING. Shifting = derivatives, works by changing the shape of the payoff.
Forward commitments vs options	Forwards: no cash up front, both obligated, outcome locked. Options: premium up front, buyer has a right only, upside retained.
Risk tolerance vs risk budgeting	Tolerance = HOW MUCH risk is acceptable. Budgeting = HOW and WHERE that risk gets taken.
Open-end vs closed-end funds	Open-end always trades at NAV. Closed-end trades at a premium or discount.
Buyout vs venture capital	Buyout: a FEW LARGE investments expected to work. VC: MANY SMALL investments, most expected to fail.
DC vs DB pension plans	DC: the EMPLOYEE bears investment and inflation risk. DB: the EMPLOYER does.

Appendix C - The volume in one page

The thread that ties the six modules together.

Volume 9 is not six topics. It is one argument, and if you can state the argument you can reconstruct most of the detail.

1. SOME RISK IS PAID FOR AND SOME IS NOT. Combining assets that do not move together lowers risk without lowering return, so any risk you could have diversified away earns nothing. Only systematic risk - measured by beta - is compensated. (Modules 1 and 2)
2. THEREFORE THE PORTFOLIO, NOT THE SECURITY, IS THE UNIT OF ANALYSIS. Judge every holding by what it contributes to the whole. Enron's employees did not, and lost their job and their savings in the same event. (Module 3)
3. BUT 'THE PORTFOLIO' MEANS ALL OF SOMEONE'S WEALTH, NOT JUST THE ACCOUNT YOU MANAGE. Human capital, pensions, property and liabilities all belong in the picture. This is why an IPS starts with the person and not the market, and why the same theory produces a 5%-bond portfolio for Yale and an 80%-bond portfolio for an insurer. (Module 4)
4. PEOPLE WILL NOT FOLLOW THE PLAN. They will hold losers, sell winners, ignore contradictory evidence, anchor on their purchase price, keep employer stock and do nothing out of inertia. Cognitive errors can be corrected with better process; emotional biases can only be designed around. (Module 5)
5. SO YOU NEED MACHINERY TO HOLD THE LINE. Decide the tolerance in advance and in writing, budget the risk across allocation, tactical bets and selection, measure exposure continuously, and modify it by avoidance, acceptance, transfer or shifting when it drifts out of line. (Module 6)

KEY IDEA

Risk is not the enemy. Unmeasured, unchosen, unmanaged risk is. Every formula, every bias and every governance structure in this volume exists to keep the risk you actually hold lined up with the risk you deliberately chose.

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